

Solutions

Maths

Q1. Let OABCDE is a regular hexagon, O is origin and position vector of A and B are $\vec{a}$ and $\vec{b}$ respectively. The position vector of point of intersection of OB and AC is :

- (A) $\frac{2}{3}\vec{b}$ (B) $\frac{2}{3}(\vec{a} + \vec{b})$ (C) $\frac{\vec{b}}{3}$ (D) $\vec{a} - \frac{\vec{b}}{3}$

Ans. A

Sol. $\vec{AB} = \vec{b} - \vec{a}$

observe that in regular hexagon

length of OC is double the length of its side and $OC \parallel AB$

$$\Rightarrow \vec{OC} = 2(\vec{b} - \vec{a})$$

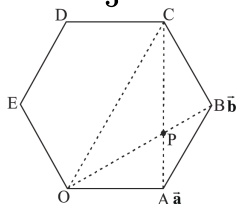
$$\text{Line OB : } \vec{r} = \lambda \vec{b}$$

$$\text{Line AC : } \vec{r} = \vec{a} + \mu(2\vec{b} - 2\vec{a} - \vec{a})$$

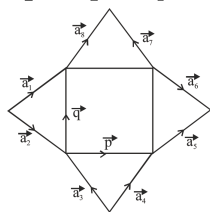
$$\text{For P } \lambda = 2\mu, 1 - 3\mu = 0$$

$$\Rightarrow \mu = \frac{1}{3}, \lambda = \frac{2}{3}$$

$$\Rightarrow \vec{r} = \frac{2}{3}\vec{b}$$



Q2. Figure shows a square. On each side of a square an outward equilateral triangle is constructed, $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4, \vec{a}_5, \vec{a}_6, \vec{a}_7, \vec{a}_8, \vec{p}$ & $\vec{q}$ are as shown in figure. If $\vec{a}_2 \times \vec{a}_1 + \vec{a}_4 \times \vec{a}_3 + \vec{a}_6 \times \vec{a}_5 + \vec{a}_8 \times \vec{a}_7 = \lambda \vec{p} \times \vec{q}$, then λ is equal to :



- (A) $\sqrt{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $2\sqrt{3}$ (D) $\frac{\sqrt{3}}{4}$

Ans. C

Sol. $\vec{a}_{i+1} \times \vec{a}_i = \frac{\sqrt{3}}{2} l^2 \hat{n}$

$$\vec{p} \times \vec{q} = l^2 \hat{n}$$

$$2\sqrt{3}l^2 \hat{n} = \lambda l^2 \hat{n}$$

$$\lambda = 2\sqrt{3}$$

($\hat{n}$ is unit vector $\perp$ to plane and ' l' ' is length of square)

- Q3. Four vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{x}$ satisfy the relation $(\vec{a} \cdot \vec{x})\vec{b} = \vec{c} + \vec{x}$ where $\vec{b} \cdot \vec{a} \neq 1$. The value of $\vec{x}$ in terms of $\vec{a}, \vec{b}$ and $\vec{c}$ is equal to :

(A) $\frac{(\vec{a} \cdot \vec{c})\vec{b} - \vec{c}(\vec{a} \cdot \vec{b} - 1)}{(\vec{a} \cdot \vec{b} - 1)}$ (B) $\frac{\vec{c}}{\vec{a} \cdot \vec{b} - 1}$ (C) $\frac{2(\vec{a} \cdot \vec{c})\vec{b} + \vec{c}}{\vec{a} \cdot \vec{b} - 1}$ (D) $\frac{2(\vec{a} \cdot \vec{c})\vec{c} + \vec{c}}{\vec{a} \cdot \vec{b} - 1}$

Ans. A

Sol. $(\vec{a} \cdot \vec{x})\vec{b} = \vec{c} + \vec{x}$ _____ (1)

taking dot product with $\vec{a}$

$$(\vec{a} \cdot \vec{x})(\vec{b} \cdot \vec{a}) = (\vec{c} \cdot \vec{a}) + (\vec{a} \cdot \vec{x})$$

$$\Rightarrow (\vec{a} \cdot \vec{x}) \cdot (\vec{b} \cdot \vec{a} - 1)$$

$$= \vec{c} \cdot \vec{a}$$

$$\Rightarrow \vec{a} \cdot \vec{x}$$

$$= \frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a} - 1} \text{ _____ (2)}$$

from (1) and (2)

$$\left(\frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a} - 1} \right) \vec{b} = \vec{c} + \vec{x}$$

$$\Rightarrow \vec{x} = \left(\frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a} - 1} \right) \vec{b} - \vec{c}$$

$$= \frac{(\vec{a} \cdot \vec{c})\vec{b} - \vec{c}(\vec{a} \cdot \vec{b} - 1)}{(\vec{a} \cdot \vec{b} - 1)}$$

- Q4. If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors and α, β, γ are the angles between the vectors $\vec{a}; \vec{b}; \vec{c}$ and $\vec{a}, \vec{b}, \vec{c}$ respectively and also α, β, γ are the angles which the unit vector $l\hat{i} + m\hat{j} + n\hat{k}$ makes with the positive x-axis, y-axis and z-axis respectively, then the value of $|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$ equals :

(A) $|\sum l - 1|$ (B) $2 \sum (lm - 1)$ (C) $-2 \sum lm + 2 \sum l + 2$ (D) $2(1 - \sum l/m)$

Ans. A

Sol. $(l + m + n)^2 = 1 + 2(ln + mn + nl)$

$$\vec{a} \cdot \vec{b} = \cos \alpha = l$$

$$\vec{b} \cdot \vec{c} = \cos \beta = m$$

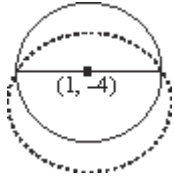
$$\vec{c} \cdot \vec{a} = \cos \gamma = n$$

$$\begin{aligned} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|^2 &= |\vec{a} \times \vec{b}|^2 + |\vec{b} \times \vec{c}|^2 + |\vec{c} \times \vec{a}|^2 + 2(\vec{a} \times \vec{b})(\vec{b} \times \vec{c}) + 2(\vec{b} \times \vec{c})(\vec{c} \times \vec{a}) + 2(\vec{c} \times \vec{a})(\vec{a} \times \vec{b}) \\ &= \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + 2[\cos \alpha \cos \beta + \cos \beta \cos \gamma + \cos \gamma \cos \alpha - \cos \alpha - \cos \beta - \cos \gamma] \\ &= 1 - \cos^2 \alpha + 1 - \cos^2 \beta + 1 - \cos^2 \gamma + 2[lm + mn + nl - l - m - n] \\ &= 3 - (l^2 + m^2 + n^2) + 2(lm + mn + nl) - 2(l + m + n) \\ &= 2 - 2(lm + mn + nl) - 2(l + m + n) \\ &= 1 + 2(lm + mn + nl) - (l + m + n)^2 \\ &= 1 + (l + m + n)^2 - 2(l + m + n) \\ &= (1 - \sum l)^2 \end{aligned}$$

- Q5. If the circle $x^2 + y^2 + 4x + 22y + c = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - d = 0$. If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{4-z}{\lambda}$ and $\frac{x-1}{\lambda} = \frac{y-4}{2} = \frac{z-\alpha}{1}$ intersect, then the value of $\alpha\lambda$ is (Where, α is equal to $\frac{c+d}{10}$):
- (A) 1 (B) 2 (C) $\frac{1}{2}$ (D) 0

Ans. D

Sol.



Common chord

$$6x + 14y + c + d = 0$$

$$(1, -4)$$

$$6 - 56 + c + d = 0$$

$$\Rightarrow c + d = 50$$

$$\lambda = \frac{c+d}{10}$$

$$\lambda = 5$$

Lines are intersecting

$$\Rightarrow (\vec{b} - \vec{a}) \cdot (\vec{P}_1 \times \vec{P}_2) = 0$$

$$\begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -\lambda \\ \lambda & 2 & 1 \end{vmatrix} = 0$$

$$(1 + 2\lambda) + (1 + \lambda^2) - (2 - \lambda) = 0$$

$$\lambda^2 + 3\lambda = 0$$

$$\Rightarrow \lambda = 0, -3$$

- Q6. If $\vec{a}$ and $\vec{b}$ are non-zero, non-collinear vectors such that $|\vec{a}| = 2$, $\vec{a} \cdot \vec{b} = 1$ and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$. If $\vec{r}$ is any vector such that $\vec{r} \cdot \vec{a} = 2$, $\vec{r} \cdot \vec{b} = 8$, $(\vec{r} + 2\vec{a} - 10\vec{b}) \cdot (\vec{a} \times \vec{b}) = 4\sqrt{3}$ and $\vec{r} + 2\vec{a} - 10\vec{b} = \lambda(\vec{a} \times \vec{b})$ then λ is equal to :

- (A) $\frac{1}{2}$ (B) 2 (C) $\frac{1}{4}$ (D) None of these

Ans. D

Sol. $\lambda(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) = 4\sqrt{3} (\vec{a} \cdot \vec{b} \cos \frac{\pi}{3} = 1)$

$$\lambda(a^2b^2 - (\vec{a} \cdot \vec{b})^2) = 4\sqrt{3}$$

$$\Rightarrow b = 1$$

$$\lambda(4 \times 1 - (1)^2) = 4\sqrt{3}$$

$$\lambda = \frac{4\sqrt{3}}{3}$$

- Q7. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors, then the value of $|\vec{a} - 2\vec{b}|^2 + |\vec{b} - 2\vec{c}|^2 + |\vec{c} - 2\vec{a}|^2$ does not exceed :

- (A) 9 (B) 12 (C) 18 (D) 21

Ans. D

$$\begin{aligned}
 \text{Sol. } & |\vec{a} - 2\vec{b}|^2 + |\vec{b} - 2\vec{c}|^2 + |\vec{c} - 2\vec{a}|^2 \\
 &= 5a^2 + 5b^2 + 5c^2 - 4(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\
 &= 15 - 4(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\
 &\leq 15 - 4\left(\frac{-3}{2}\right) \\
 &= 21 \\
 \therefore \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} &\geq \frac{-3}{2}
 \end{aligned}$$

Q8. The adjacent side vectors $\vec{OA}$ and $\vec{OB}$ of a rectangle OACB are $\vec{a}$ and $\vec{b}$ respectively where O is the origin. If $16|\vec{a} \times \vec{b}| = 3(|\vec{a}| + |\vec{b}|)^2$ and θ be the acute angle between the diagonals OC and AB then the possible value of $\tan(\theta/2)$ is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{3}$

Ans. D

$$\begin{aligned}
 \text{Sol. } & 16|\vec{a}||\vec{b}|\sin\frac{\pi}{2} = 3|\vec{a}|^2 + 3|\vec{b}|^2 + 6|\vec{a}||\vec{b}| \\
 &\Rightarrow 3a^2 - 10ab + 3b^2 = 0 \\
 &\Rightarrow (3a - b)(a - 3b) = 0 \\
 \text{Now } & \vec{OC} \cdot \vec{AB} = (\vec{a} + \vec{b}) \cdot (\vec{b} - \vec{a}) \\
 &= |\vec{OC}||\vec{AB}|\cos\theta \\
 &\Rightarrow \frac{b^2 - a^2}{\sqrt{a^2 + b^2}\sqrt{a^2 + b^2}} = \cos\theta \\
 &= \frac{9a^2 - a^2}{9a^2 + a^2} \text{ \{using } b = 3a\} \\
 \therefore \cos\theta &= \frac{4}{5} \\
 \therefore \tan\theta/2 &= \sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}} \\
 &= \sqrt{\frac{1 - 4/5}{1 + 4/5}} \\
 &= \frac{1}{3}
 \end{aligned}$$

Q9. If $\vec{d} = \lambda(\vec{a} \times \vec{b}) + \mu(\vec{b} \times \vec{c}) + \nu(\vec{c} \times \vec{a})$ $[\vec{a} \ \vec{b} \ \vec{c}] = \frac{1}{8}$ and $\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = 8$, then $\lambda + \mu + \nu$ is :

(A) 32 (B) 64 (C) 27 (D) 19

Ans. B

$$\begin{aligned}
 \text{Sol. } & \vec{d} \cdot \vec{c} = \lambda[\vec{a} \ \vec{b} \ \vec{c}] + \mu \cdot 0 + \nu \cdot 0 \\
 & \vec{d} \cdot \vec{a} = \mu[\vec{a} \ \vec{b} \ \vec{c}] \\
 & \vec{d} \cdot \vec{b} = \nu[\vec{a} \ \vec{b} \ \vec{c}] \\
 & \vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = [\vec{a} \ \vec{b} \ \vec{c}](\lambda + \mu + \nu) \\
 & \lambda + \mu + \nu = 64
 \end{aligned}$$

- Q10. If $[\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] + [\vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a}] = 8$, then area of tetrahedron OABC with three of its sides along $\vec{a}, \vec{b}, \vec{c}$ can be :
- (A) $\frac{2}{3}$ (B) $\frac{1}{6}$ (C) $\frac{5}{6}$ (D) $\frac{1}{8}$

Ans. A

Sol. $[abc]^2 + 2[abc] = 8$
 $\Rightarrow [abc] = 2, -4$
 Area = $\frac{1}{3}$ or $\frac{2}{3}$

- Q11. Let A, B, C are three points in x-y plane such that $OA = 2, OB = 2, OC = 3$ and $AB^2 + BC^2 + CA^2 = 51$, then the distance between O and D is (where D is mid-point of AB and O is origin) :

- (A) $\frac{1}{2}$ (B) 2 (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. C

Sol. Let vector $\vec{a}, \vec{b}, \vec{c}$ are positions vectors of points A, B, C respectively then form the given condition

$$\vec{a} + \vec{b} + \vec{c} = 0$$

So, $OD = \frac{OC}{2}$

- Q12. Points A, B lie on $x + 2y + z - 10 = 0$ and $\frac{x-5}{2} = \frac{y-2}{1} = \frac{z-1}{-2}$ respectively. C, D divide OA, OB in ratio 1 : 2, 1 : 3 respectively (O is origin). Locus of C is P and locus of D is L. Point of intersection of P and L is:

- (A) $\left(\frac{21}{12}, \frac{11}{12}, \frac{1}{12}\right)$ (B) $\left(\frac{25}{12}, \frac{11}{12}, \frac{-7}{12}\right)$ (C) $\left(\frac{27}{12}, \frac{17}{12}, \frac{-5}{12}\right)$ (D) None of these

Ans. B

Sol. P : $x + 2y + z = \frac{10}{3}$
 $\frac{x - \frac{5}{4}}{2} = \frac{y - \frac{1}{2}}{1} = \frac{z - \frac{1}{4}}{-2}$
 L :

For point of intersection $\left(2t + \frac{5}{4}\right) + 2\left(t + \frac{1}{2}\right) + \left(-2t + \frac{1}{4}\right) = \frac{10}{3}$

$$2t + \frac{10}{4} = \frac{10}{3}$$

Hence, point of intersection is $\left(\frac{25}{12}, \frac{11}{12}, \frac{-7}{12}\right)$

- Q13. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors with magnitude 2, 3 and 1 respectively and $\vec{a} \cdot \vec{c} = \frac{1}{4}$. If the following relation holds true for some $\lambda (\neq 0)$, $\vec{b} - \lambda \vec{a} = 3\vec{c}$, then the value of λ is :

- (A) $\frac{8}{3}$ (B) $-\frac{8}{3}$ (C) $\frac{3}{8}$ (D) $-\frac{3}{8}$

Ans. D

Sol. $\vec{a} \cdot (\vec{b} - 3\vec{c}) = \vec{a} \cdot (\lambda\vec{a})$
 $\Rightarrow \vec{a} \cdot \vec{b} - 3\vec{a} \cdot \vec{c}$
 $= \lambda|\vec{a}|^2$
 $\Rightarrow \vec{a} \cdot \vec{b} - 3\frac{3}{4} = 4\lambda$
 $\Rightarrow \vec{a} \cdot \vec{b} = 4\lambda + \frac{3}{4} \dots\dots\dots(1)$
 $\vec{b} \cdot (\vec{b} - 3\vec{c}) = \vec{b} \cdot (\lambda\vec{a})$
 $\Rightarrow |\vec{b}|^2 - 3\vec{b} \cdot \vec{c}$
 $= \lambda(\vec{a} \cdot \vec{b})$
 $\Rightarrow 9 - 3\vec{b} \cdot \vec{c}$
 $= \lambda\left(4\lambda + \frac{3}{4}\right)$ using equation (1)
 $\Rightarrow 3\vec{b} \cdot \vec{c}$
 $= 9 - \lambda\left(4\lambda + \frac{3}{4}\right) \dots\dots\dots(2)$
 $\vec{c} \cdot (\vec{b} - 3\vec{c}) = \vec{c} \cdot (\lambda\vec{a})$
 $\Rightarrow \vec{b} \cdot \vec{c} - 3|\vec{c}|^2 = \frac{\lambda}{4}$
 $\Rightarrow 3 - \frac{\lambda}{3}\left(4\lambda + \frac{3}{4}\right) - 3$ using equation (2)
 $= \frac{\lambda}{4}$
 $\Rightarrow -\frac{\lambda}{3}\left(4\lambda + \frac{3}{4}\right)$
 $= \frac{\lambda}{4}$
 $\Rightarrow 4\lambda + \frac{3}{4}$
 $= -\frac{3}{4}$
 $\Rightarrow 4\lambda$
 $= -\frac{3}{2}$
 $\Rightarrow \lambda$
 $= -\frac{3}{8}$

Q14. The vectors $\vec{x}$ and $\vec{y}$ satisfy the equation $p\vec{x} + q\vec{y} = \vec{a}$ (where p, q are scalar constants and $\vec{a}$ is a known vector). It is given that $\vec{x} \cdot \vec{y} \geq \frac{|\vec{a}|^2}{4pq}$, then $\frac{|\vec{x}|}{|\vec{y}|}$ is equal to ($pq > 0$)

- (A) 1 (B) $\frac{p^2}{q^2}$ (C) $\frac{p}{q}$ (D) $\frac{q}{p}$

Ans. D

Sol. $p\vec{x} + q\vec{y} = \vec{a}$
Let $p\vec{x} - q\vec{y} = R\hat{n}$
 $R = |p\vec{x} - q\vec{y}|$
 $|p\vec{x} - q\vec{y}|^2 = |p\vec{x} + q\vec{y}|^2 - 4pq\vec{x} \cdot \vec{y}$ but $\vec{x} \cdot \vec{y} \geq \frac{|\vec{a}|^2}{4pq}$
 ≥ 0

$$\begin{aligned}
&\Rightarrow |\vec{a}|^2 \\
&= 4pq\vec{x} \cdot \vec{y} \\
&\Rightarrow |p\vec{x} \cdot q\vec{y}| \\
&= 0 \\
&\Rightarrow p\vec{x} \\
&= q\vec{y} \\
&\Rightarrow \vec{x} \\
&= \frac{\vec{a}}{2p}, \vec{y} \\
&= \frac{\vec{a}}{2p} \\
\frac{|\vec{x}|}{|\vec{y}|} &= \frac{q}{p}
\end{aligned}$$

Q15. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three mutually perpendicular unit vectors and $\vec{d}$ is a unit vector which makes equal angle with $\vec{a}, \vec{b}$ and $\vec{c}$, then $|\vec{a} + \vec{b} + \vec{c} + \vec{d}|^2$ is equal to

- (A) 4 (B) $4 \pm \sqrt{3}$ (C) $4 \pm 2\sqrt{3}$ (D) none of these

Ans. C

Sol. $|\vec{a} + \vec{b} + \vec{c} + \vec{d}|^2 = \Sigma |\vec{a}|^2 + 2\Sigma \vec{a} \cdot \vec{b}$
 $= 4 + 2\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c})$
 $\vec{d} \cdot \vec{a} = \vec{d} \cdot \vec{b}$
Let $\vec{d} = \lambda\vec{a} + \mu\vec{b} + \nu\vec{c}$, then $\vec{d} \cdot \vec{c} = \cos \theta$

$$\begin{aligned}
\Rightarrow \lambda = \mu = \nu &= \cos \theta \\
\lambda^2 + \mu^2 + \nu^2 &= 1 \\
&\Rightarrow 3 \cos^2 \theta \\
&= 1
\end{aligned}$$

Also $\Rightarrow \cos \theta = \pm \frac{1}{\sqrt{3}}$

$$\begin{aligned}
\therefore |\vec{a} + \vec{b} + \vec{c} + \vec{d}|^2 &= 4 \pm 2 \frac{3}{\sqrt{3}} \\
&= 4 \pm 2\sqrt{3}
\end{aligned}$$

$$\begin{aligned}
\lambda^2 + \mu^2 + \nu^2 &= 1 \\
&\Rightarrow 3 \cos^2 \theta \\
&= 1 \\
&\Rightarrow \cos \theta = \pm \frac{1}{\sqrt{3}} \\
\therefore |\vec{a} + \vec{b} + \vec{c} + \vec{d}|^2 &= 4 \pm 2 \frac{3}{\sqrt{3}} \\
&= 4 \pm 2\sqrt{3}
\end{aligned}$$

Q16. Given vectors $\vec{a} = \sin \frac{\pi}{6} \hat{i} + \sin \frac{\pi}{3} \hat{j}$, $\vec{b} = \sin \frac{\pi}{3} \hat{j} + \sin \frac{\pi}{6} \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$. The volume of parallelepiped with three conterminous edges as $\vec{u} = (\vec{a} \cdot \vec{a})\vec{a} + (\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c})\vec{c}$, $\vec{v} = (\vec{a} \cdot \vec{b})\vec{a} + (\vec{b} \cdot \vec{b})\vec{b} + (\vec{b} \cdot \vec{c})\vec{c}$, $\vec{w} = (\vec{a} \cdot \vec{c})\vec{a} + (\vec{b} \cdot \vec{c})\vec{b} + (\vec{c} \cdot \vec{c})\vec{c}$ equals :

- (A) $\left(2 \cos \frac{\pi}{4} - \sin \frac{\pi}{3}\right)^3$ (B) $\left(2 \cos \frac{\pi}{3} - \sin \frac{\pi}{3}\right)^3$ (C) $\left(2 \cos \frac{\pi}{6} - \sin \frac{\pi}{6}\right)^3$
 (D) $\left(2 \sin \frac{\pi}{4} - \cos \frac{\pi}{3}\right)^3$

Ans. C

Sol.

$$[\vec{u}\vec{v}\vec{w}] = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} [\vec{a}\vec{b}\vec{c}]$$

$$= [\vec{a}\vec{b}\vec{c}]^3$$

$$[\vec{a}\vec{b}\vec{c}] = \begin{vmatrix} 1/2 & \sqrt{3}/2 & 0 \\ 0 & \sqrt{3}/2 & 1/2 \\ 0 & 2 & 3 \end{vmatrix} = \frac{1}{2} \left(\frac{3\sqrt{3}}{2} - 1 \right)$$

Q17. Let $\vec{a}$ and $\vec{b}$ be two unit vectors. The maximum value of $\frac{|\vec{a} + \vec{b}|^2 - |\vec{a} - \vec{b}|^2}{|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2}$ is :

- (A) 1/2 (B) 1 (C) 2 (D) 4

Ans. B

Sol. Let θ = angle between $\vec{a}$ and $\vec{b}$.

We have

$$|\vec{a} + \vec{b}|^2 = 2 + 2 \cos \theta = 4 \cos^2 \frac{\theta}{2}$$

$$\text{and } |\vec{a} - \vec{b}|^2 = 2 - 2 \cos \theta = 4 \sin^2 \frac{\theta}{2}$$

$$\therefore \frac{|\vec{a} + \vec{b}|^2 - |\vec{a} - \vec{b}|^2}{|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2} = \frac{4 \cos \theta}{4} = \cos \theta$$

Clearly, maximum value = 1. (which is possible when $\theta = 0$.)

Q18. If $\vec{q}, \vec{r}$ are unit vectors such that $\vec{p} = \vec{q} \times \vec{p} + \vec{r}$, then :

- (A) $\vec{p} \cdot \vec{q} = \vec{r} \cdot \vec{q}$ (B) maximum value of $[\vec{p}\vec{q}\vec{r}] = \frac{1}{2}$ (C) minimum value of $[\vec{p}\vec{q}\vec{r}] = 0$
 (D) minimum value of $[\vec{p}\vec{q}\vec{r}] = -1$

Ans. A, B, C

Sol. Take dot product with $\vec{q}$ both the sides

$$\vec{p} \cdot \vec{q} = \vec{r} \cdot \vec{p}$$

$$\vec{p} - \vec{q} \times \vec{p} = \vec{r}$$

$$\text{Squaring } p^2 + p^2 \sin^2 \theta = 1$$

$$p^2 = \frac{1}{1 + \sin^2 \theta}$$

$$\vec{p} = \vec{q} \times \vec{p} + \vec{r}$$

$$\text{Squaring } p^2 = p^2 \sin^2 \theta + 2[\vec{q}\vec{p}\vec{r}] + 1$$

$$\begin{aligned}
2[\vec{p}\vec{q}\vec{r}] &= 1 - p^2 \cos^2 \theta \\
&= 1 - \frac{\cos^2 \theta}{(1 + \sin^2 \theta)} \\
2[\vec{p}\vec{q}\vec{r}] &= 1 - \frac{(1 - \sin^2 \theta)}{(1 + \sin^2 \theta)} \\
&= \frac{2 \sin^2 \theta}{(1 + \sin^2 \theta)} \\
2[\vec{p}\vec{q}\vec{r}] &= 2 \left(1 - \frac{1}{1 + \sin^2 \theta} \right) \\
[\vec{p}\vec{q}\vec{r}] &= 1 - \frac{1}{(1 + \sin^2 \theta)} \\
\text{When } \sin^2 \theta = 0 &\Rightarrow [\vec{p}\vec{q}\vec{r}] = 0 \\
\text{When } \sin^2 \theta = 1 &\Rightarrow [\vec{p}\vec{q}\vec{r}] = \frac{1}{2}
\end{aligned}$$

Q19. Let $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}| = |\vec{b}| = |\vec{c}| = 2$ and $\vec{a} \wedge \vec{b} = \vec{b} \wedge \vec{c} = \vec{c} \wedge \vec{a} = \frac{\pi}{3}$, then :

- (A) volume of parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ as three coterminous edges is $8\sqrt{2}$
- (B) volume of parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ as three coterminous edges is $4\sqrt{2}$
- (C) height of parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ as three coterminous edges is $\frac{2\sqrt{2}}{\sqrt{3}}$ whose base is a parallelogram with adjacent sides $\vec{a}, \vec{b}$.
- (D) height of parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ as three coterminous edges is $\frac{4\sqrt{2}}{\sqrt{3}}$ whose base is a parallelogram with adjacent sides $\vec{a}, \vec{b}$.

Ans. B, C

Sol.

$$\begin{aligned}
[\vec{a}\vec{b}\vec{c}]^2 &= \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} \\
[\vec{a}\vec{b}\vec{c}]^2 &= \begin{vmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{vmatrix} = 4(12) - 2(4) + 2(-4) = 32 \\
v &= 4\sqrt{2}
\end{aligned}$$

$$v = (\text{base area}) (\text{ht}) \Rightarrow 4\sqrt{2} = 4 \times \frac{\sqrt{3}}{2} \times \text{ht}$$

$$\therefore \text{base area} = \text{area of parallelogram} = \left| \vec{a} \times \vec{b} \right| = 2 \cdot 2 \cdot \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{2\sqrt{2}}{\sqrt{3}} = \text{ht}$$

Q20. The volume of the parallelepiped whose coterminous edges are represented by the vectors $2\vec{b} \times \vec{c}$,

$$3\vec{c} \times \vec{a} \text{ and } 4\vec{a} \times \vec{b} \text{ where } \vec{a} = (1 + \sin \theta)\hat{j} + \sin 2\theta\hat{k},$$

$$\vec{b} = \sin \left(\theta + \frac{2\pi}{3} \right) \hat{i} + \cos \left(\theta + \frac{2\pi}{3} \right) \hat{j} + \sin \left(2\theta + \frac{4\pi}{3} \right) \hat{k},$$

$$\vec{c} = \sin \left(\theta - \frac{2\pi}{3} \right) \hat{i} + \cos \left(\theta - \frac{2\pi}{3} \right) \hat{j} + \sin \left(2\theta - \frac{4\pi}{3} \right) \hat{k} \text{ is 18 cubic units, then the values of } \theta, \text{ in}$$

the interval

$\left(0, \frac{\pi}{2}\right)$, is/are :

- (A) $\frac{\pi}{9}$ (B) $\frac{2\pi}{9}$ (C) $\frac{\pi}{3}$ (D) $\frac{4\pi}{9}$

Ans. A, B, D

Sol. Volume = $\begin{vmatrix} 4\vec{a} \times \vec{b} & 2\vec{b} \times \vec{c} & 3\vec{c} \times \vec{a} \end{vmatrix}$
= 18

$24[\vec{a} \quad \vec{b} \quad \vec{c}] = 18$

$$\begin{vmatrix} 0 & 1 + \sin \theta & \sin 2\theta \\ \sin \left(\theta + \frac{2\pi}{3}\right) & \cos \left(\theta + \frac{2\pi}{3}\right) & \sin \left(2\theta + \frac{4\pi}{3}\right) \\ \sin \left(\theta - \frac{2\pi}{3}\right) & \cos \left(\theta - \frac{2\pi}{3}\right) & \sin \left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix} = \frac{3}{4}$$

$$-(1 + \sin \theta) \left[\sin \left(\theta + \frac{2\pi}{3}\right) \sin \left(2\theta - \frac{4\pi}{3}\right) - \sin \left(2\theta + \frac{4\pi}{3}\right) \sin \left(\theta - \frac{2\pi}{3}\right) \right]$$

$$+ \sin 2\theta \left[\sin \left(\theta + \frac{2\pi}{3}\right) \cos \left(\theta - \frac{2\pi}{3}\right) - \cos \left(\theta + \frac{2\pi}{3}\right) \sin \left(\theta - \frac{2\pi}{3}\right) \right] = \pm \frac{\sqrt{3}}{2}$$

$$\frac{(1 + \sin \theta)}{2} \left[\cos \left(3\theta - \frac{2\pi}{3}\right) - \cos(2\pi - \theta) - \cos \left(3\theta + \frac{2\pi}{3}\right) + \cos(\theta - 2\pi) \right]$$

$$+ \sin 2\theta \left(\frac{\sqrt{3}}{2} \right) = \pm \frac{\sqrt{3}}{2}$$

$$\frac{1 + \sin \theta}{2} (-2 \sin 3\theta \sin \frac{2\pi}{3}) + (\sin 2\theta) \left(\frac{\sqrt{3}}{2} \right) = \pm \frac{\sqrt{3}}{2}$$

$$\sin 3\theta(1 + \sin \theta) - \sin 2\theta = \pm 1$$

Q21. If $\vec{a}, \vec{b}, \vec{c}$ are non-zero, non-collinear vectors such that a vector $\vec{p} = ab \cos(2\pi - (\vec{a} \wedge \vec{b}))\vec{c}$ and a vector $\vec{q} = ac \cos(\pi - (\vec{a} \wedge \vec{c}))\vec{b}$ then $\vec{p} + \vec{q}$ is :

- (A) parallel to $\vec{a}$ (B) perpendicular to $\vec{a}$ (C) coplanar with $\vec{b}$ and $\vec{c}$ (D) coplanar with $\vec{a}$ and $\vec{c}$

Ans. B, C

Sol. $\vec{p} = ab \cos(2\pi - (\vec{a} \wedge \vec{b}))\vec{c}$
= $(\vec{a} \times \vec{b}) \cdot \vec{c}$

$\vec{q} = ac \cos(\pi - (\vec{a} \wedge \vec{c}))\vec{b}$

$\vec{q} = (\vec{a} \times \vec{c}) \cdot \vec{b}$

Now

$$\vec{a} \cdot (\vec{p} + \vec{q}) = \vec{a} \cdot \vec{p} + \vec{a} \cdot \vec{q}$$

$$= \vec{a} \cdot (\vec{a} \times \vec{b})\vec{c} + \vec{a} \cdot (\vec{a} \times \vec{c})\vec{b}$$

$$= [\vec{a} \vec{a} \vec{b}]\vec{c} + [\vec{a} \vec{a} \vec{c}]\vec{b}$$

but $[\vec{a} \vec{a} \vec{b}] = [\vec{a} \vec{a} \vec{c}] = 0$

$$= 0 + 0 = 0$$

Q22. Three points A, B and C with position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are collinear if :

(A) $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 0$

(B) $\exists$ a unique value of λ for which the point $C(\vec{c})$ satisfies the vector equation $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

(C) the area of the $\triangle ABC$ vanishes.

(D) $\exists$ scalars α, β, γ not all zero simultaneously such that $\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = 0$, where $\alpha + \beta + \gamma = 0$.

Ans. A, B, C, D

Sol. If $\vec{a}, \vec{b}, \vec{c}$ are collinear vectors, then $\vec{c} - \vec{a} = \lambda(\vec{b} - \vec{a})$. Area of triangle

$$= \frac{1}{2}(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$$

$$= 0$$

Q23. Let $\vec{a}, \vec{b}$ be two unit vectors and $\vec{r}$ be a vector such that $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} + \vec{b})$.

If $\vec{a} \cdot \vec{b} = \frac{1}{2}$, $\vec{r} \cdot \vec{a} = 2$, $\vec{r} \cdot \vec{b} = 1$ and $|\vec{r}| = \vec{r} \cdot (\vec{a} \times \vec{b}) = 2$, then :

(A) $x = 1$ (B) $y = 1$ (C) $z = 1$ (D) $x^2 + y^2 + z^2 = 2$

Ans. A, C, D

Sol. $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} + \vec{b})$

$$\vec{r} \cdot \vec{a} = 2 = x + \frac{y}{2} \dots\dots\dots(1)$$

$$\vec{r} \cdot \vec{b} = 1 = \frac{x}{2} + y \dots\dots\dots(2)$$

$$\vec{r} \cdot \vec{r} = 4 = 2x + y + 2z \dots\dots\dots(3)$$

From (1), (2) and (3) $x = z = 1$ and $y = 0$.

$$|\vec{r}| = \vec{r} \cdot (\vec{a} \times \vec{b}) = 2$$

$$|\vec{r}|^2 = (\vec{r} \cdot (\vec{a} \times \vec{b}))^2$$

$$(\vec{r} \cdot \vec{a})^2 (\vec{r} \cdot \vec{b})^2 - (\vec{r} \cdot \vec{a} \cdot \vec{b})^2 = 4$$

$$(2)^2 (1)^2 - (\vec{r} \cdot \vec{a} \cdot \vec{b})^2 = 4$$

$$4 - (\vec{r} \cdot \vec{a} \cdot \vec{b})^2 = 4$$

$$\vec{r} \cdot \vec{a} \cdot \vec{b} = 0$$

$$x = 1, y = 0, z = 1$$

$$\vec{r} = \vec{a} + \vec{a} + \vec{b} = 2\vec{a} + \vec{b}$$

$$\vec{r} \cdot \vec{a} = 2\vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{a}$$

$$= 2(1) + \frac{1}{2}$$

$$= \frac{5}{2}$$

But given $\vec{r} \cdot \vec{a} = 2$

So, option (A) is incorrect.

$$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$$

$$\vec{r} \cdot \vec{a} = x + \frac{y}{2} = 2$$

$$\vec{r} \cdot \vec{b} = \frac{x}{2} + y = 1$$

$$x = 2, y = 0$$

$$|\vec{r}|^2 = x^2 + y^2 + z^2 = 4$$

$$4 + 0 + z^2 = 4$$

$$z = 0$$

$$\vec{r} = 2\vec{a}$$

$$\vec{r} \cdot (\vec{a} \times \vec{b}) = 2\vec{a} \cdot (\vec{a} \times \vec{b})$$

$$= 0$$

But given $\vec{r} \cdot (\vec{a} \times \vec{b}) = 2$

So, option (B) is incorrect.

$$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} + \vec{b})$$

$$\vec{r} \cdot \vec{a} = x + \frac{y}{2} + z(1 + \frac{1}{2}) = 2$$

$$x + \frac{y}{2} + \frac{3z}{2} = 2 \dots\dots\dots(1)$$

$$\vec{r} \cdot \vec{b} = \frac{x}{2} + y + z(\frac{1}{2} + 1) = 1$$

$$\frac{x}{2} + y + \frac{3z}{2} = 1 \dots\dots\dots(2)$$

$$|\vec{r}|^2 = x^2 + y^2 + z^2 + xy + xz + yz = 4$$

Subtracting (2) from (1)

$$\frac{x}{2} - \frac{y}{2} = 1$$

$$x - y = 2$$

Adding (1) and (2)

$$\frac{3x}{2} + \frac{3y}{2} + 3z = 3$$

$$x + y + 2z = 2$$

$$x = 2 + y$$

$$2 + y + y + 2z = 2$$

$$2y + 2z = 0$$

$$y = -z$$

$$x = 2 - z$$

$$|\vec{r}|^2 = (2 - z)^2 + (-z)^2 + z^2 + (2 - z)(-z) + (2 - z)z + (-z)z = 4$$

$$4 - 4z + z^2 + z^2 + z^2 - 2z + z^2 + 2z - z^2 - z^2 = 4$$

$$z^2 - 4z = 0$$

$$z(z - 4) = 0$$

$$z = 0 \text{ or } z = 4$$

If $z = 0$, then $y = 0, x = 2$

$$\vec{r} = 2\vec{a}$$

$$\vec{r} \cdot (\vec{a} \times \vec{b}) = 2\vec{a} \cdot (\vec{a} \times \vec{b}) = 0$$

But given $\vec{r} \cdot (\vec{a} \times \vec{b}) = 2$

So, option (C) is incorrect.

$$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$$

$$\vec{r} \cdot \vec{a} = x + \frac{y}{2} = 2$$

$$\vec{r} \cdot \vec{b} = \frac{x}{2} + y = 1$$

$$x = 2, y = 0$$

$$|\vec{r}|^2 = x^2 + y^2 + z^2 = 4$$

$$4 + 0 + z^2 = 4$$

$$z = 0$$

$$\vec{r} = 2\vec{a}$$

$$\vec{r} \cdot (\vec{a} \times \vec{b}) = 2\vec{a} \cdot (\vec{a} \times \vec{b}) = 0$$

But given $\vec{r} \cdot (\vec{a} \times \vec{b}) = 2$

So, option (D) is incorrect.

Q24. A pyramid with vertex at the point P, whose position vector is $4\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$ has a regular hexagonal base ABCDEF. Position vectors of points A and B are $\hat{i}$ and $\hat{i} + 2\hat{j}$ respectively. Centre of the base has the position vector $\hat{i} + \hat{j} + \sqrt{3}\hat{k}$. Altitude drawn from P on the base meets the diagonal AD at point G and volume of the pyramid is $6\sqrt{3}$ cubic units then all possible position vectors of G are :

- (A) $\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$ (B) $-(\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k})$ (C) $3\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$ (D) $-3\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$

Ans. B, C

Sol. Let the centre of the base be (O)

$$\vec{AB} = 2\hat{i} \Rightarrow |\vec{AB}| = 2$$

$$\text{Area of the } \triangle OAB = \frac{1}{4} \cdot 4 \cdot \sqrt{3} = \sqrt{3}$$

$$\Rightarrow \text{Base area} = 6\sqrt{3} \text{ sq. units}$$

Let height of the pyramid be h.

$$\Rightarrow \frac{1}{3} \cdot 6\sqrt{3}h = 6\sqrt{3} \Rightarrow h = 3 \text{ units}$$

$$\vec{AP} = 3\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$$

$$|\vec{AP}| = 5 \text{ units}, |\vec{AG}| = 4 \text{ units}$$

Now, $\vec{AG}$ and $\vec{AO}$ are collinear

$$\vec{AG} = \lambda \vec{AO}$$

$$= |\vec{AG}|$$

$$= |\lambda| |\vec{AO}|$$

$$\Rightarrow 2|\lambda| = 4 \Rightarrow |\lambda| = 2$$

$$\vec{AG} = \pm 2(\hat{i} + \hat{j} + \sqrt{3}\hat{k})$$

$$\vec{G} = \pm 2(\hat{i} + \hat{j} + \sqrt{3}\hat{k}) + \hat{i}$$

$$= 2\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k} + \hat{i}$$

$$= -2\hat{i} - 2\hat{j} - 2\sqrt{3}\hat{k} + \hat{i}$$

$$= 3\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}, -\hat{i} - 2\hat{j} - 2\sqrt{3}\hat{k}$$

Q25. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + \hat{k}, \vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}, \vec{c} \times \vec{b} = \vec{d} \times \vec{b}$ and $\vec{d} \cdot \vec{a} = 8$ then which of the following is/are correct :

- (A) $\vec{d} \cdot \vec{b} = 4$ (B) $\vec{d} \cdot \vec{c} = 14$ (C) $[\vec{b}\vec{c}\vec{d}] = 0$ (D) $[\vec{a}\vec{b}\vec{c}] = 2$

Ans. B, C, D

$$\text{Sol. } [\vec{a}\vec{b}\vec{c}] = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 3 & 1 \end{vmatrix} = 1(-1-3) - 1(1-2) + 1(3+2) = -4 + 1 + 5 = 2$$

$$\vec{c} \times \vec{b} = \vec{d} \times \vec{b}$$

$$\Rightarrow (\vec{c} - \vec{d}) \times \vec{b}$$

$$= 0$$

$$\begin{aligned}
&\Rightarrow \vec{b}, \vec{c} \text{ and } \vec{d} \text{ are coplanar.} \\
&\Rightarrow \vec{c} \\
&= \vec{d} + \lambda \vec{b} \\
&\Rightarrow \vec{c} \cdot \vec{c} \\
&= \vec{d} \cdot \vec{c} + 0 \\
&= 14
\end{aligned}$$

Q26. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors each of magnitude $\sqrt{3}$ and the angle between each pair of them is $\pi/3$, If $\vec{b} \times \vec{c} + \vec{c} \times \vec{a} = x\vec{a} + y\vec{b} + z\vec{c}$ where x, y, z are scalars, then :

(A) Volume of parallelepiped formed by $\vec{a}, \vec{b}, \vec{c}$ is $\frac{3\sqrt{3}}{\sqrt{2}}$ (B) The value of x is $\frac{\sqrt{3}}{\sqrt{2}}$

(C) $x^2 + y^2 + z^2 \leq \sqrt{21}$

(D) Number of integral co-ordinates (x, y, z) satisfying $x^2 + y^2 + z^2 \leq 12$ is 125

Ans. A, C

Sol. $|\vec{a}| = |\vec{b}| = |\vec{c}| = \sqrt{3}$

$$\begin{aligned}
\vec{a} \wedge \vec{b} &= \vec{a} \wedge \vec{c} \\
&= \vec{b} \wedge \vec{c} \\
&= 60^\circ
\end{aligned}$$

$$\vec{b} \times \vec{c} + \vec{c} \times \vec{a} = x\vec{a} + y\vec{b} + z\vec{c}$$

(A) $[\vec{a}\vec{b}\vec{c}] = 3x + \frac{y}{2} + \frac{z}{2}$

(B) $[\vec{a}\vec{b}\vec{c}] = \frac{3}{2}x + 3y + \frac{3}{2}z$

(C) $0 \Rightarrow \frac{3}{2}x + \frac{3}{2}y + 3z$

$$\Rightarrow x = y = -z$$

$$\begin{vmatrix}
\vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\
(\text{Volume})^2 \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\
\vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c}
\end{vmatrix}
= \begin{vmatrix}
3 & 3/2 & 3/2 \\
3/2 & 3 & 3/2 \\
3/2 & 3/2 & 3
\end{vmatrix} = \frac{27}{2}$$

$$= \frac{3\sqrt{3}}{\sqrt{2}}$$

$$\Rightarrow x$$

$$= \frac{\sqrt{3}}{\sqrt{2}}$$

$$x^2 + y^2 + z^2 = 3x^2 = \frac{3 \cdot 3}{2} = \frac{9}{2} < \sqrt{21}$$

Q27. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three distinct non-zero vectors such that $\vec{a} \times \vec{b} = -\vec{b} \times \vec{c}$ then which of the following is/are correct

(A) $|(\vec{a} + \vec{c}) \cdot \vec{b}| = |\vec{a} + \vec{c}| |\vec{b}|$ (B) $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$ (C) $(\vec{a} + \vec{b}) \cdot [(\vec{b} + \vec{c}) \times (\vec{c} + \vec{a})] = 2[\vec{a}\vec{b}\vec{c}]$

(D) $\vec{a} = \lambda \vec{b} + \mu \vec{c} (\lambda, \mu \in R)$

Ans. B, C, D

Sol. $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = 0$
 $\Rightarrow (\vec{a} - \vec{c}) \times \vec{b} = 0$
 $\Rightarrow \vec{a} - \vec{c} = \lambda \vec{b}$
 $\Rightarrow \vec{a} = \lambda \vec{b} + \vec{c}$
 $\Rightarrow \vec{a}, \vec{b}, \vec{c}$ are coplanar.

Q28. From point $A(-1, 1, -1)$, the perpendicular is drawn to a plane P_1 whose foot is $(1, -1, 1)$. From point $B(3, -1, -1)$, the perpendicular is drawn to a plane P_2 whose foot is $(1, 2, 1)$. P_3 is a plane containing the line of intersection of P_1 and P_2 and passing through the mid point of the line joining A and B. Then which of the following statement (s) is/are correct :

(A) Direction ratios of line of intersection of P_1 and P_2 are 5, 4, -1.

(B) Equation of plane P_3 is $16x - 19y + 4z = 12$.

(C) Equation of line of intersection of P_1 and P_2 is $\frac{x-15}{5} = \frac{y-12}{4} = \frac{z-0}{-1}$

(D) The length of perpendicular from $(0, 0, 0)$ to plane P_3 is $\frac{2}{\sqrt{18}}$

Ans. A, B, C

Sol. Plane $P_1 = (2x - 1) - 2(y + 1) + 2(z - 1) = 0 \Rightarrow x - y + z = 3$

Plane $P_2 = 2(x - 1) - 3(y - 2) - 2(z - 1) = 0$

$\Rightarrow 2x - 3y - 2z + 6 = 0$

so $P_1 + \lambda P_2 = 0 \Rightarrow (x - y + z - 3) + \lambda(2x - 3y - 2z + 6) = 0$

$(1 - 1 - 3) + \lambda(2 + 2 + 6) = 0$

$\lambda = \frac{3}{10}$

$10(x - y + z - 3) + 3(2x - 3y - 2z + 6) = 0$

$16x - 19y + 4z - 12 = 0$

put $z = 0$ in P_1 and P_2

$x - y = 3, 2x - 3y = -6$

By solving we get $x = 15, y = 12, z = 0$

Q29. $a_1, a_2, a_3 \in R - \{0\}$ and $a_1 + a_2 \cos^2 x + a_3 \sin^2 x = 0 \forall x \in R$, then which of the following is/are true :

(A) $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ and $\vec{b} = 4\hat{i} + 2\hat{j} + \hat{k}$ are perpendicular. $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$

(B) $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} + 2\hat{k}$ are parallel. $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$

(C) If $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ is of length $\sqrt{6}$ units, then (a_1, a_2, a_3) can be $(1, -1, -2)$

(D) If $2a_1 + 3a_2 + 6a_3 = 26$ then $|a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}| = 2\sqrt{6}$ units.
 $a_1, a_2, a_3 \in R - \{0\}$

Ans. A, B, C, D

Sol. $(a_1 + a_2) + \sin^2 x(a_3 - 2a_2) = 0$

$\Rightarrow a_1 + a_2 = 0$ and $a_3 - 2a_2 = 0$

$\Rightarrow \frac{a_1}{-1} = \frac{a_2}{1} = \frac{a_3}{2} = \lambda$

$$a_1 = -\lambda, a_2 = \lambda, a_3 = 2\lambda$$

$$\vec{a} \cdot \vec{b} = -4\lambda + 2\lambda + 2\lambda = 0$$

$$\vec{a} = \lambda \cdot \vec{b} \Rightarrow \vec{a} \text{ is parallel to } \vec{b}$$

Q30. The position vector of the vertices A, B & C of a tetrahedron ABCD are $(1, 1, 1)$, $(1, 0, 0)$ & $(3, 0, 0)$ respectively. The altitude from the vertex D to the opposite face ABC meets the median through A of $\triangle ABC$ at point E. If $AD = 4$ units and volume of tetrahedron $= \frac{2\sqrt{2}}{3}$, then the correct statement(s) among the following is/are :

- (A) The altitude from vertex D = 2 units (B) There is only one possible position for point E
(C) There are two possible positions for point E (D) Vector $\hat{j} - \hat{k}$ is normal to the plane ABC.

Ans. A, C, D

Sol. Volume आयतन $= \frac{2\sqrt{2}}{3}$

$$\Rightarrow \frac{1}{3} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & -1 & -1 \end{vmatrix} \times h = \frac{2\sqrt{2}}{3}$$

$$\Rightarrow h|\hat{j} - \hat{k}| = 2\sqrt{2} \Rightarrow h = 2$$

for E, let AE : EM = $\lambda : 1$

$$\Rightarrow E = \left(\frac{2\lambda + 1}{\lambda + 1}, \frac{1}{\lambda + 1}, \frac{1}{\lambda + 1} \right) \text{ and } (AE)^2 + (ED)^2 = (AD)^2$$

$$\Rightarrow \lambda = -2 \text{ or } \lambda = -2/3$$

$$E = (3, -1, -1) \text{ or } E = (-1, 3, 3)$$



Q31. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three non-coplanar vectors, then $[\vec{a} \times (\vec{b} + \vec{c})\vec{b} \times (\vec{c} - 2\vec{a})\vec{c} \times (\vec{a} + 3\vec{b})]$ is equal to

(A) $[\vec{a}\vec{b}\vec{c}]^2$ (B) $7[\vec{a}\vec{b}\vec{c}]^2$ (C) $-5[\vec{a} \times \vec{b}\vec{b} \times \vec{c}\vec{c} \times \vec{a}]$ (D) $7[\vec{c} \times \vec{a}\vec{a} \times \vec{b}\vec{b} \times \vec{c}]$

Ans. B, D

Sol. $[\vec{a} \times \vec{b} - \vec{c} \times \vec{a}\vec{b} \times \vec{c} + 2\vec{a} \times \vec{b}\vec{c} \times \vec{a} - 3\vec{b} \times \vec{c}]$

$$\begin{vmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 0 & -3 & 1 \end{vmatrix} [\vec{a} \times \vec{b}\vec{b} \times \vec{c}\vec{c} \times \vec{a}] = 7[\vec{c} \times \vec{a}\vec{a} \times \vec{b}\vec{b} \times \vec{c}]$$

Q32. If unit vectors $\hat{i}$ and $\hat{j}$ are at right angles to each other and $\vec{p} = 3\hat{i} + 4\hat{j}$, $\vec{q} = 5\hat{i}$, $4\vec{r} = \vec{p} + \vec{q}$ and $2\vec{s} = \vec{p} - \vec{q}$, then :

- (A) $|\vec{r} + k\vec{s}| = |\vec{r} - k\vec{s}|$ for all real k (B) $\vec{r}$ is perpendicular to $\vec{s}$
(C) $\vec{r} + \vec{s}$ is perpendicular to $\vec{r} - \vec{s}$ (D) $|\vec{r}| = |\vec{s}| = |\vec{p}| = |\vec{q}|$

Ans. A, B, C

Sol. We have $\hat{i} \cdot \hat{j} = 0$

given $\vec{p} = 3\hat{i} + 4\hat{j}$, $\vec{q} = 5\hat{i}$

$$4\vec{r} = \vec{p} + \vec{q}, 2\vec{s} = \vec{p} - \vec{q}$$

$$\Rightarrow \vec{r} = 2\hat{i} + \hat{j}, \vec{s} = -\hat{i} + 2\hat{j}$$

for option A :

$$\begin{aligned} |\vec{r} + k\vec{s}| &= |(2-k)\hat{i} + (1+2k)\hat{j}| \\ &= \sqrt{(2-k)^2 + (1+2k)^2} \\ &= \sqrt{5+5k^2} \dots\dots\dots (i) \\ \text{and } |\vec{r} - k\vec{s}| &= |(2+k)\hat{i} + (1-2k)\hat{j}| \\ &= \sqrt{(2+k)^2 + (1-2k)^2} \\ &= \sqrt{5+5k^2} \dots\dots\dots (ii) \end{aligned}$$

from (i) and (ii)

$$|\vec{r} + k\vec{s}| = |\vec{r} - k\vec{s}|$$

option B :

$$\vec{r} \cdot \vec{s} = 0$$

$$\vec{r} \perp \vec{s}$$

option C :

$$\begin{aligned} \vec{r} + \vec{s} &= \hat{i} + 3\hat{j} \text{ and } \vec{r} - \vec{s} \\ &= 3\hat{i} - \hat{j} \end{aligned}$$

$$(\vec{r} + \vec{s}) \cdot (\vec{r} - \vec{s}) = 0$$

Hence $(\vec{r} + \vec{s})$ is perpendicular to $(\vec{r} - \vec{s})$

option D :

$$\begin{aligned} |\vec{r}| &= \sqrt{5}, |\vec{s}| \\ &= \sqrt{5}, |\vec{p}| \\ &= 5, |\vec{q}| \\ &= 5 \quad \therefore |\vec{r}| \\ &= |\vec{s}| \neq |\vec{p}| \\ &= |\vec{q}| \end{aligned}$$

Q33. The vectors $\vec{a}(x) = (\cos x)\hat{i} + (\sin x)\hat{j}$, $\vec{b}(x) = x\hat{i} + (\sin x)\hat{j}$ are collinear for :

- (A) unique value of x , $0 < x < \frac{\pi}{6}$ (B) unique value of x , $\frac{\pi}{6} < x < \frac{\pi}{3}$
 (C) no value of x , $0 < x < \frac{\pi}{6}$ (D) infinitely many values of x , $0 < x < \frac{\pi}{2}$

Ans. B, C

Sol. $\vec{a}(x)$ and $\vec{b}(x)$ are collinear if and only if $\cos x = x$,

$$\text{now let } f(x) = x - \cos x, f'(x) = 1 + \sin x \geq 0$$

$f(x)$ is increasing and hence $f(x) = 0$ for a unique value of x

For $x \geq \frac{\pi}{3}$ $f(x) > 0$ and for $x < \frac{\pi}{6}$, $f(x) < 0$ thus $\cos x = x$, for unique value $x \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

$|\vec{OA}| = \ell$, $|\vec{OB}| = m$, $|\vec{OC}| = n$. O is origin A, B, C lie in the plane $x + 2y - z = 0$ and

$|\vec{AB}|^2 + |\vec{BC}|^2 + |\vec{CA}|^2$ is maximum

Q34. If $\vec{p}, \vec{q}, \vec{r}$ are three non-coplanar vectors satisfying

$$x(-\vec{p} + 2\vec{q} + 3\vec{r}) + y(2\vec{p} + 3\vec{q} - 4\vec{r}) + z(\vec{p} - 4\vec{q} + 5\vec{r}) = -4\vec{p} - 21\vec{q} + 38\vec{r}, \text{ then choose the correct}$$

option (s) :

- (A) $5x - y + z = 17$ (B) $2x - 2y + 6z = 33$ (C) $x + 5y - 3z = 25$ (D) $4x + y + 2z = 13$

Ans. A, D

Sol. -

Q35. Given a line L and a plane $\Pi : L = \frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}, \Pi = x + y + z = 3$. Let lines L_1 and L_2 respectively be projection and image of line L in plane Π . Let planes Π_1, Π_2 respectively be planes containing L and parallel, perpendicular to Π . Then shortest distance of :

- (A) Π_1 from origin is $\sqrt{3}$ (B) L_1 from $(-9, 6, -3)$ is $\sqrt{53}$ (C) L_2 from origin is $\sqrt{\frac{285}{7}}$
 (D) Π_2 from origin is $\sqrt{\frac{49}{26}}$

Ans. A, B, D

Sol. -

Paragraph

The vertices of $\triangle ABC$ are $A(2, 0, 0), B(0, 1, 0), C(0, 0, 2)$. Its orthocentre is H and circumcentre is S . P is a point equidistant from A, B, C and the origin O .

Q36. The z -coordinate of H is :

- (A) 1 (B) $1/2$ (C) $1/6$ (D) $1/3$

Ans. D

Sol. $AB = BC$

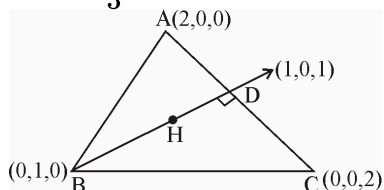
$$\text{p.v. of } H = \hat{j} + r(\hat{i} - \hat{j} + \hat{k})$$

$$\text{Also } \overrightarrow{AH} \cdot \overrightarrow{BC} = 0$$

$$\Rightarrow [(r-2)\hat{i} + (1-r)\hat{j} + r\hat{k}] \cdot (-\hat{j} + 2\hat{k}) = 0$$

$$\Rightarrow r - 1 + 2r = 0$$

$$\Rightarrow r = \frac{1}{3}$$



Q37. The y -coordinate of S is :

- (A) $5/6$ (B) $1/3$ (C) $1/6$ (D) $1/2$

Ans. C

Sol.

$$\text{p.v. of } H = \frac{\hat{i}}{3} + \frac{2\hat{j}}{3} + \frac{\hat{k}}{3}$$

$$\text{p.v. of centroid} = \frac{2\hat{i}}{3} + \frac{\hat{j}}{3} + \frac{2\hat{k}}{3}$$

$$\text{p.v. of } S = \frac{3(\text{p.v. of centroid}) - \text{p.v. of } H}{2}$$

$$y \text{ coordinate of } S = \frac{1}{6}$$

Q38. PA :

- (A) 1 (B) $\sqrt{2}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{3}{2}$

Ans. D

Sol. Let $P = (a, b, c)$

$$\Rightarrow PA = PB = PC = OP$$

$$\Rightarrow (a-2)^2 + b^2 + c^2 = a^2 + (b-1)^2 + c^2$$

$$= a^2 + b^2 + (c-2)^2 = a^2 + b^2 + c^2$$

$$\Rightarrow P = \left(1, \frac{1}{2}, 1\right)$$

$$PA = \frac{3}{2}$$

Paragraph

If $\vec{x} \times \vec{y} = \vec{a}$ _____ (i)

$\vec{y} \times \vec{z} = \vec{b}$ _____ (ii)

$\vec{x} \cdot \vec{b} = r$ _____ (iii)

$\vec{x} \cdot \vec{y} = 1$ _____ (iv) and $\vec{y} \cdot \vec{z} = 1$ _____ (v),

Where $\vec{a}, \vec{b}$ are non-zero non-collinear vectors, then.

Q39. Vector $\vec{x}$ is :

- (A) $\frac{1}{|\vec{a} \times \vec{b}|^2} [\vec{a} \times (\vec{a} \times \vec{b})]$ (B) $\frac{r}{|\vec{a} \times \vec{b}|} [\vec{a} \times \vec{b} - \vec{a} \times (\vec{a} \times \vec{b})]$ (C) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} \times \vec{b} + \vec{b} \times (\vec{a} \times \vec{b})]$
(D) None of these

Ans. B

Sol. -

Q40. Vector $\vec{y}$ is :

- (A) $\frac{\vec{a} \times \vec{b}}{r}$ (B) $\vec{a} + \frac{\vec{a} \times \vec{b}}{r}$ (C) $\frac{\vec{a} \times \vec{b}}{2r}$ (D) $\frac{\vec{a}}{r} + \vec{a} \times \vec{b}$

Ans. A

Sol. -

Q41. $\vec{z}$ is :

- (A) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} + \vec{b} + (\vec{a} \times \vec{b})]$ (B) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} + \vec{b} - \vec{a} \times (\vec{a} \times \vec{b})]$
(C) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} \times \vec{b} + \vec{b} \times (\vec{a} \times \vec{b})]$ (D) None of these

Ans. C

Sol. (i) and (ii) give

$$(\vec{x} \times \vec{y}) \times (\vec{y} \times \vec{z}) = (\vec{a} \times \vec{b})$$

$$[\vec{x}\vec{y}\vec{z}]\vec{y} = \vec{a} \times \vec{b}$$

$$\text{also (ii) gives } \vec{x} \cdot (\vec{y} \times \vec{z}) = \vec{x} \cdot \vec{b}$$

$$= r$$

$$\Rightarrow \vec{y}$$

$$= \frac{\vec{a} \times \vec{b}}{r}$$

$$\text{thus, (i) gives } (\vec{x} \times \vec{y}) \times \vec{y} = \frac{\vec{a} \times (\vec{a} \times \vec{b})}{r}$$

$$\vec{y} - |\vec{y}|^2 \vec{x} = -\frac{\vec{a} \times (\vec{a} \times \vec{b})}{r}$$

$$\frac{\vec{a} \times \vec{b}}{r} - \frac{|\vec{a} \times \vec{b}|^2}{r^2} \vec{x} = -\frac{\vec{a} \times (\vec{a} \times \vec{b})}{r}$$

similarly, take cross with $\vec{y}$ in equation (ii)

इसी प्रकार समीकरण (ii) में $\vec{y}$ के साथ सदिश गुणन से

$$(\vec{y} \times \vec{z}) \times \vec{y} = |\vec{y}|^2 \vec{z} (\vec{y} \cdot \vec{z}) \vec{y}$$

$$\frac{\vec{b} \times (\vec{a} \times \vec{b})}{r} = \frac{|\vec{a} \times \vec{b}|^2}{r^2} \vec{z} - \frac{\vec{a} \times \vec{b}}{r}$$

Paragraph

A conic section is obtained by the intersection of two inverted cones (having same vertex) with a plane. If a plane passes through the vertex, then its intersection with the cones either represents a pair of straight lines or a point depending upon whether it intersects the cones or not. If plane does not pass through vertex, then section may be hyperbola or circle depending upon whether plane is parallel to axis of the cone or perpendicular to it. If it has any other inclination, then intersection of plane and cone gives an ellipse. If a variable point P moves such that the line passing through P and Q (0, 0, 2) makes an angle 60° with z-axis, then

Q42. The locus of P is :

- (A) $x^2 - y^2 - 3(z - 2)^2 = 0$ (B) $x^2 - y^2 + 3(z - 2)^2 = 0$ (C) $x^2 + y^2 - 3(z - 2)^2 = 0$
 (D) $x^2 + y^2 + 3(z - 2)^2 = 0$

Ans. C

Sol. P (x, y, z) lies on line through (0, 0, 2) making 60° with z-axis

$$\text{i.e. } \frac{x\hat{i} + y\hat{j} + (z - 2)\hat{k}}{\sqrt{x^2 + y^2 + (z - 2)^2}} = \cos 60^\circ$$

$$x^2 + y^2 - 3(z - 2)^2 = 0$$

Q43. The locus of intersection of locus of P and the plane $x + y + z = 2$ is :

- (A) a pair of straight lines (B) a hyperbola (C) a circle (D) a point

Ans. A

Sol. The plane passes through (0, 0, 2) and its inclination with z-axis is α , where $\cos \alpha = \frac{1}{\sqrt{3}}$

i.e. $\alpha < 60^\circ$

it represent a pair of straight lines.

Q44. The locus of intersection of locus of P with $x + y = 2$ is :

- (A) a straight line (B) a hyperbola (C) a circle (D) an ellipse

Ans. B

Sol. The plane is parallel to z-axis
" locus is a hyperbola

Q45. Three vectors $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$ having magnitude 2, 3 and $5^{1/3}$ respectively satisfying $\vec{\gamma} \times ((\vec{\alpha} \times \vec{\gamma}) \times \vec{\gamma}) = \vec{\beta}$, then $\sin(\vec{\alpha} \wedge \vec{\gamma})$ is equal to _____.

Ans. 0.30

Sol. $|\vec{\alpha}| = 2, |\vec{\beta}| = 3, |\vec{\gamma}| = 5^{1/3}$
 $\vec{\gamma} \times ((\vec{\alpha} \times \vec{\gamma}) \times \vec{\gamma}) = \vec{\beta}$
 $\vec{\gamma} \perp (\vec{\alpha} \times \vec{\gamma}) \times \vec{\gamma}$, take magnitude
 $5^{1/3} |(\vec{\alpha} \times \vec{\gamma}) \times \vec{\gamma}| = 3$
 $(\vec{\alpha} \times \vec{\gamma}) \perp \vec{\gamma}$
 $|\vec{\alpha} \times \vec{\gamma}| \cdot 5^{2/3} = 3$
 $2 \cdot 5^{1/3} \cdot 5^{1/3} \cdot \sin(\vec{\alpha} \wedge \vec{\gamma}) = 3$
 $\sin(\vec{\alpha} \wedge \vec{\gamma}) = \frac{3}{10}$

Q46. Let P be a plane containing the line $z = 2, x = -y$ and passing through a point $M(\alpha, \beta, \gamma)$ where $\alpha > 0, \beta > 0, \gamma > 0$. The distance of M from yz plane, zx plane and xy plane are 1, 2 and 3 respectively. Plane P meets co-ordinate axes at points A, B and C respectively. Find the volume of tetrahedron OABC (O is the origin).

Ans. 12

Sol. $\frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{0}$.
P(1, -1, 2), M(1, 2, 3)
DR's of AB : (0, -3, -1)
 $\hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -3 & -1 \\ 1 & -1 & 0 \end{vmatrix} = -\hat{i} - \hat{j} + 3\hat{k}$
 $< -1, -1, 3 >$
 $-(x-1) - (y-2) + 3(z-3) = 0$
 $-x - y + 3z + 1 + 2 - 9 = 0$
 $x + y - 3z = -6$
pts $(-6, 0, 0), (0, -6, 0)$ and $(0, 0, 2)$
Volume = $\frac{1}{6} |\vec{OA} \quad \vec{OB} \quad \vec{OC}|$
 $= \frac{1}{6} \begin{vmatrix} -6 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 2 \end{vmatrix}$
 $= \frac{1}{6} \times 6 \times 12 = 12$

Q47. Let $\pi : x + y - z - 4 = 0$ be the equation of a plane and A be the point with position vector $\hat{i} + 2\hat{j} - 3\hat{k}$. L is a line which passes through the point (1, 2, 3) with direction ratios 3, -1 and 4. If the distance of the point A from the line L measured parallel to the plane π is d_1 and the distance of the point A from the plane π measured parallel to the line L is d_2 , then find the value of $\sqrt{d_1^2 - d_2^2}$.

Ans. 10

Sol. Plane $\pi : x + y - z = 0$

$$\vec{OA} = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\text{Eqn. L: } \frac{x-1}{3} = \frac{y-2}{-1} = \frac{z-3}{4} = \lambda \text{ (let)}$$

Point $P(3\lambda + 1, -\lambda + 2, 4\lambda + 3)$ lies on the line.

Point $A(1, 2, -3)$

DRs. of line from A and P

$$(3\lambda, -\lambda, 4\lambda - 6)$$

$$3\lambda - \lambda - 4\lambda - 6 = 0$$

$$-2\lambda - 6 = 0$$

$$\lambda = -3$$

$$AB = d_1 = \sqrt{9\lambda^2 + \lambda^2 + 16\lambda^2 + 36 + 48\lambda}$$

$$d_1 = \sqrt{126}$$

$$\Rightarrow \text{equ. of } \parallel \text{ line } \frac{x-1}{3} = \frac{y-2}{-1} = \frac{z+3}{4} = \lambda$$

$$x = 3\lambda + 1, y = 2 - \lambda, z = 4\lambda - 3$$

$$AC : (3\lambda, -\lambda, 4\lambda)$$

$$3\lambda + 1 - \lambda + 2 + 2 - (4\lambda - 3) = 4$$

$$2\lambda - 4\lambda + 6 = 4$$

$$-2\lambda = -2$$

$$\lambda = 1$$

$$AC = \sqrt{(9 + 1 + 16)\lambda^2}$$

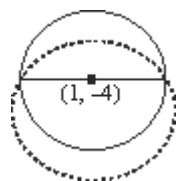
$$d_2 = \sqrt{26 \times 1} = \sqrt{26}$$

$$\sqrt{d_1^2 - d_2^2} = \sqrt{126 - 26} = 10$$

Q48. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where $a, b > 0$), then the maximum value of (ab) is λ . The plane denoted by $\Pi_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\Pi_2 : 5x + 3y + 10z = \sqrt{\lambda}$. If the plane in its new position be denoted by Π , and the distance of this plane from the origin is $\sqrt{k}$ where $k \in N$, then find k

Ans. 212

Sol.



$$S_1 - S_2 = 0$$

$$6x + 14y + a + b = 0$$

$$(1, -4)$$

$$6 - 56 + a + b = 0$$

$$a + b = 50$$

$$AM \geq GM$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

$$ab \leq 25^2$$

$$ab \leq 625$$

$$\lambda = 625$$

Let new plane be $P_1 + \lambda P_2 = 0$

$$(4x + 7y + 4z + 81) + \lambda(5x + 3y + 10z - 25) = 0$$

$$(5\lambda + 4)x + (3\lambda + 7)y + (10\lambda + 4)z + 81 - 25\lambda = 0$$

perpendicular to P_1

$$\Rightarrow 4(5\lambda + 4) + 7(3\lambda + 7) + 4(10\lambda + 4) = 0$$

$$81\lambda = -81$$

$$\lambda = -1$$

New plane be $-x + 4y - 6z + 106 = 0$

$$(0, 0, 0)$$

$$\begin{aligned} P &= \frac{106}{\sqrt{1 + 16 + 36}} \\ &= \sqrt{\frac{106 \times 106}{53}} \\ &= \sqrt{212} \end{aligned}$$

Q49. If $\vec{a}, \vec{b}$ and $\vec{c}$ are mutually perpendicular vectors such that $|\vec{a}| = \sqrt{6}, |\vec{b}| = \sqrt{2}, |\vec{c}| = \sqrt{3}$ and a vector $\vec{v} = (\vec{a} \times (\vec{a} \times \vec{b})) \times \vec{c} + (\vec{b} \times (\vec{b} \times \vec{c})) \times \vec{a} + (\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b}$ where magnitude of $\vec{v}$ is M then $\frac{M^2}{100}$ is _____.

Ans. 3.96

Sol. Sol. $\therefore (\vec{a} \times (\vec{a} \times \vec{b})) \times \vec{c} = ((\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b}) \times \vec{c}$

$$= -|\vec{a}|^2 \vec{b} \times \vec{c} = -6(\vec{b} \times \vec{c})$$

$$\text{Similarly } (\vec{b} \times (\vec{b} \times \vec{c})) \times \vec{a} = -|\vec{b}|^2 \vec{c} \times \vec{a}$$

$$= -2(\vec{c} \times \vec{a}) \text{ and } (\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b} = -|\vec{c}|^2 \vec{a} \times \vec{b} = -3(\vec{a} \times \vec{b})$$

$$\text{Now, } \vec{v} = -[3(\vec{a} \times \vec{b}) + 6(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})]$$

$\therefore \vec{a}, \vec{b}$ and $\vec{c}$ are mutually perpendicular

$\therefore \vec{a} \times \vec{b}, \vec{b} \times \vec{c}$ and $\vec{c} \times \vec{a}$ are also mutually perpendicular

$$\begin{aligned} \Rightarrow |\vec{v}| &= \sqrt{9|\vec{a}|^2|\vec{b}|^2 + 36|\vec{b}|^2|\vec{c}|^2 + 4|\vec{c}|^2|\vec{a}|^2} \\ &= 6\sqrt{11} \end{aligned}$$

$$M = 6\sqrt{11}$$

$$M^2 = 36 \times 11$$

$$\frac{M^2}{100} = \frac{3.96}{100}$$

Q50. Consider a line passing through the point P(3, 4, 5) & having direction ratios (2, -2, 1). If image of point Q(1, -1, 8) w.r.t. given line is Q', then length of median of $\Delta PQQ'$ through vertex P is _____.

Ans. 3.00

Sol. $\Delta PQQ'$ is an isosceles triangle and $\vec{PM} = 2\hat{i} - 2\hat{j} + \hat{k}$

$$\text{Now, } PM = |\vec{PQ}| \cos \theta, \vec{PQ} = 2\hat{i} + 5\hat{j} - 3\hat{k}$$

$$\begin{aligned}
 PM &= \left| \frac{\vec{PQ} \cdot \vec{PM}}{|\vec{PM}|} \right| \\
 &= \left| \frac{4 - 10 - 3}{3} \right| \\
 &= 3
 \end{aligned}$$

Q51. If lines $bx = 1 - cy$, $bz = 1$ and $bx = 2 + ay$, $bz = 3 + cy$ are perpendicular, where a, b, c are sides of ΔABC , then minimum value of $\cos B$ is _____.

Ans. 0.5

Sol. lines are $\frac{x - \frac{1}{b}}{c} = \frac{y - \frac{1}{b}}{-b} = \frac{z - \frac{1}{b}}{0}$ and $\frac{x - \frac{2}{a}}{a} = \frac{y - \frac{1}{b}}{b} = \frac{z - \frac{3}{c}}{c}$ are perpendicular

$$\Rightarrow ac - b^2 = 0 \Rightarrow b^2 = ac$$

Now,

$$\begin{aligned}
 \cos B &= \frac{a^2 + c^2 - b^2}{2ac} \\
 &= \frac{1}{2} \left(\frac{a}{c} + \frac{c}{a} \right) - \frac{1}{2} \\
 &\geq \frac{1}{2}(2) - \frac{1}{2} \left(\because x + \frac{1}{x} \geq 2, x > 0 \right)
 \end{aligned}$$

$$\Rightarrow \text{minimum value of } \cos B = \frac{1}{2} \text{ exist when } a = c$$

Q52. Let $|\vec{a}| = 4$, $|\vec{b}| = 1$ and $\vec{b} \times \vec{a} + \vec{a} = \vec{a}$. If projection of $\vec{a}$ along $\vec{b}$ is 3 then projection of $\vec{a}$ along $\vec{a}$ is

Ans. 3.12

Sol. Sol.

$$\vec{b} \cdot (\vec{b} \times \vec{a} + \vec{a}) = \vec{b} \cdot \vec{a} \quad (1)$$

$$\therefore \text{projection of } \vec{a} \text{ on } \vec{b} = \vec{a} \cdot \hat{b}$$

$$\Rightarrow \vec{a} \cdot \vec{b}$$

$$= 3$$

$$\Rightarrow \vec{a} \cdot \vec{b}$$

$$= 3(\because |\vec{b}|$$

$$= 1)$$

$$\therefore \vec{b} \cdot \vec{a} = 3 \text{ (from (1))}$$

$$\text{Now } \vec{b} \times (\vec{b} \times \vec{a} + \vec{a}) = \vec{b} \times \vec{a}$$

$$\Rightarrow (\vec{b} \cdot \vec{a})\vec{b} - \vec{b}^2\vec{a} + \vec{b} \times \vec{a}$$

$$= \vec{b} \times \vec{a}$$

$$\Rightarrow 3\vec{b} - \vec{a} + \vec{b} \times \vec{a}$$

$$= \vec{b} \times \vec{a}$$

$$\Rightarrow 3\vec{b} - \vec{a} + \vec{b} \times \vec{a}$$

$$= \vec{a} - \vec{a}$$

$$\Rightarrow \vec{a}$$

$$= \frac{1}{2}(3\vec{b} + \vec{a} + \vec{b} \times \vec{a})$$

Projection of $\vec{a}$ along $\vec{a}$

$$\begin{aligned}
&= \vec{a} \cdot \hat{a} \\
&= \frac{1}{4} \left[\frac{1}{8} (3\vec{b} + \vec{a} + \vec{b} \times \vec{a}) \cdot \vec{a} \right] \\
&= \frac{1}{8} [3\vec{b} \cdot \vec{a} + |\vec{a}|^2] \\
&= \frac{1}{8} [3 \times 3 + 16] = \frac{25}{8} = 3.125
\end{aligned}$$

Q53. If $\triangle ABC$ and $\triangle AEF$ are such that $EB = BF$, $AB = EF = 1$, $BC = 6$, $AC = \sqrt{33}$ and $\vec{AB} \cdot \vec{AE} + \vec{AC} \cdot \vec{AF} = 2$, then the value of $\vec{BC} \cdot \vec{BF}$ is (given figure)

Ans. 0.85

Sol. Let M being origin then one possible equation of face LNO will be

$$x + \frac{y}{2} + \frac{z}{3} = 1$$

$$\therefore \text{required altitude} = \text{distance of LNO from M} = \frac{6}{7}$$



Q54. In a tetrahedron LMNO edges ML, MN and MO are mutually perpendicular and length of altitudes drawn from O, L and N to opposite faces are 1, 2 and 3 units respectively, then length of altitude drawn from M to face LNO is _____.

Ans. 0.85

Sol. Let M being origin then one possible equation of face LNO will be $\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1$
 $\Rightarrow x + \frac{y}{2} + \frac{z}{3} = 1 \therefore \text{required altitude} = \text{distance of LNO from M} = \frac{6}{7}$

Q55. If $\vec{a}_1 = a\hat{i} + b\hat{j}$, $\vec{a}_2 = \vec{a}_1 - (2\hat{i} + \sqrt{5}\hat{j})$ and $\vec{a}_3 = \vec{a}_1 + (2\hat{i} - \sqrt{5}\hat{j})$, then
 $\begin{vmatrix} 1 + \vec{a}_1 \cdot \vec{a}_1 & 1 + \vec{a}_1 \cdot \vec{a}_2 & 1 + \vec{a}_1 \cdot \vec{a}_3 \\ 1 + \vec{a}_2 \cdot \vec{a}_1 & 1 + \vec{a}_2 \cdot \vec{a}_2 & 1 + \vec{a}_2 \cdot \vec{a}_3 \\ 1 + \vec{a}_3 \cdot \vec{a}_1 & 1 + \vec{a}_3 \cdot \vec{a}_2 & 1 + \vec{a}_3 \cdot \vec{a}_3 \end{vmatrix}$ is equal to _____.

Ans. 80

Sol. $\vec{a}_1 = (a, b)$, $\vec{a}_2 = (a - 2, b - \sqrt{5})$, $\vec{a}_3 = (a + 2, b - \sqrt{5})$

$$\text{Area of } \Delta, \text{ whose vertices are } \vec{a}_1, \vec{a}_2 \text{ and } \vec{a}_3, \text{ is } \Delta = \frac{1}{2} \begin{vmatrix} 1 & a & b \\ 1 & a - 2 & b - \sqrt{5} \\ 1 & a + 2 & b - \sqrt{5} \end{vmatrix}$$

$$\Rightarrow \Delta^2 = \frac{1}{4} \begin{vmatrix} 1 & a & b \\ 1 & a - 2 & b - \sqrt{5} \\ 1 & a + 2 & b - \sqrt{5} \end{vmatrix} \begin{vmatrix} 1 & a - 2 & b - \sqrt{5} \\ 1 & a + 2 & b - \sqrt{5} \end{vmatrix} \text{ and } \begin{matrix} |\vec{a}_1 - \vec{a}_2| = 3, |\vec{a}_1 - \vec{a}_3| \\ = 3, |\vec{a}_2 - \vec{a}_3| \\ = 4 \end{matrix}$$

$$\Rightarrow \Delta = \sqrt{5 \times 2 \times 2 \times 1} = 2\sqrt{5}$$

$$(2\sqrt{5})^2 = \frac{1}{4} \begin{vmatrix} 1 + \vec{a}_1 \cdot \vec{a}_1 & 1 + \vec{a}_1 \cdot \vec{a}_2 & 1 + \vec{a}_1 \cdot \vec{a}_3 \\ 1 + \vec{a}_2 \cdot \vec{a}_1 & 1 + \vec{a}_2 \cdot \vec{a}_2 & 1 + \vec{a}_2 \cdot \vec{a}_3 \\ 1 + \vec{a}_3 \cdot \vec{a}_1 & 1 + \vec{a}_3 \cdot \vec{a}_2 & 1 + \vec{a}_3 \cdot \vec{a}_3 \end{vmatrix}$$

$$\therefore \text{Required answer} = 4 \times 5 \times 4 = 80$$

Q56. The volume of a tetrahedron OABC formed by the vectors $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ and $\vec{OC} = \vec{c}$ where

$|\vec{a}| = 3$, $|\vec{b}| = 4$, $|\vec{c}| = 6$, angle between the vector pairs $\{\vec{a}, \vec{b}\}$; $\{\vec{b}, \vec{a}\}$ and $\{\vec{c}, \vec{a}\}$ be $\frac{\pi}{3}$, $\cos^{-1} \frac{1}{4}$ and

$\cos^{-1}\left(\frac{-1}{18}\right)$ respectively is V. If the value of V^2 can be written as $\frac{pq}{r^2}$ where p, q, r are prime numbers then evaluate $\frac{p+q}{r}$.

Ans. 30

Sol.

$$\text{Sol. } [\vec{a}\vec{b}\vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 9 & 6 & -1 \\ 6 & 16 & 6 \\ -1 & 6 & 36 \end{vmatrix}$$

$$= 9(576 - 36) - 6(216 + 6) - (36 + 16)$$

$$= 4860 - 1332 - 52 = 3476$$

$$\therefore \text{Volume of tetrahedron } V = \frac{1}{6} |[\vec{a}\vec{b}\vec{c}]|$$

$$\therefore V^2 = \frac{1}{36} [\vec{a}\vec{b}\vec{c}]^2$$

$$= \frac{1}{36} \times 3476$$

$$= \frac{869}{9} = \frac{11 \times 79}{3^2} = \frac{pq}{r^2}$$

$$\therefore \frac{(p+q)}{r} = \frac{11+79}{3} = 30$$

Q57. A sphere is inscribed in a tetrahedron whose vertices are $A = (6, 0, 0)$, $B = (0, 4, 0)$, $C = (0, 0, 2)$ and $D = (0, 0, 0)$. If the radius of sphere is $\frac{m}{n}$ (where m and n are relatively prime), then $(m+n)$ is _____.

Ans. 5

Sol. If S_1, S_2, S_3, S_4 are areas of four faces of tetrahedron

$$\Rightarrow \frac{1}{3} r(S_1 + S_2 + S_3 + S_4) = V$$

(where V is volume of tetrahedron)

$$\therefore r = \frac{3V}{S_1 + S_2 + S_3 + S_4}$$

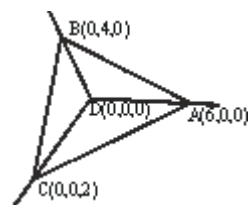
Now,

$$\therefore S_1 = 12, S_2 = 4,$$

$$S_3 = 6, S_4 = 14 \text{ and } V = \frac{1}{6} \times 48 = 8$$

$$r = \frac{24}{36} = \frac{2}{3}$$

$$\therefore m+n = 5$$



Q58. If α, β, γ & x, y, z are non-zero real numbers such that

$(\alpha x + \beta y + \gamma z) + \sqrt{(\alpha^2 + \beta^2 + \gamma^2)(x^2 + y^2 + z^2)} = 0$, then the value of $\sqrt{\frac{\alpha y}{\beta x}} + \sqrt{\frac{\beta z}{\gamma y}}$ is equal to _____.

Ans. 2

Sol. Let $\vec{p} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ and $\vec{q} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore$ Given condition

$$\Rightarrow \vec{p} \cdot \vec{q}$$

$$= -|\vec{p}||\vec{q}|$$

$$\Rightarrow \cos \theta$$

$$= -1$$

$$\Rightarrow \theta$$

$$= \pi$$

$\therefore$ vector $\vec{p}$ and $\vec{q}$ are antiparallel

$$\therefore \frac{\alpha}{x} = \frac{\beta}{y} = \frac{\gamma}{z} = k \text{ (say)}$$

$$\Rightarrow \alpha = kx, \beta = ky, \gamma = kz$$

$$\text{then } \sqrt{\frac{\alpha y}{\beta x}} + \sqrt{\frac{\beta z}{\gamma y}} = 1 + 1 = 2.$$

Q59. Let L_1 be the projection line of a line $L : \frac{x-1}{4} = \frac{y+1}{1} = \frac{z}{3}$ on the plane $\Pi : 3x - 4z - 2 = 0$. P and Q are the points on L_1 corresponding to $A = (1, -1, 0)$ and $B = (5, 0, 3)$ on the line L which divides AB externally in the ratio 31 : 5 then area of $\Delta PQR = \sqrt{\frac{p}{q}}$ where $p, q \in N$. Find the least value of $|4p - q|$

Ans. 2

Sol. Line L is parallel to the plane.

L_1 is the projection line on the plane and P and Q are foot of the parallel on the plane.

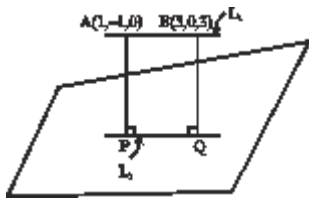
R is a point on the line L

$$\text{Area of } \Delta PQR = \frac{1}{2} \times PQ \times AP$$

$$= \frac{1}{2} \times \sqrt{26} \times \frac{1}{5} \sqrt{\frac{26}{100}}$$

$$= \sqrt{\frac{13}{50}}$$

$$\therefore 4p - q = 52 - 50 = 2$$



Q60. Number of points having positive integral co-ordinate lying on the plane $x + 2y + 3z = 15$ is n, then $\frac{n}{2}$ is equal to _____.

Ans. 6

Sol. $x + 2y + 3z = 15$

Put $z = 1$	$z = 2$	$z = 3$	$z = 4$
$x + 2y = 12$	$x + 2y = 9$	$x + 2y = 6$	$x + 2y = 3$
(10,1,1)	$x = 7, y = 1$	$x = 4, y = 1$	(1,1,4)
(8,2,1)	(7,1,2)	$x = 2, y = 2$	
(6,3,1)	(5,2,2)	(4,1,3)	

$$\begin{array}{lll} (4,4,1) & (3,3,2) & (2,2,3) \\ (2,5,1) & (1,4,2) & \\ n = 12 & & \end{array}$$

Q61. Let OA, OB, OC be coterminous edges of a cuboid. If l, m, n be the shortest distances between the sides OA, OB, OC and their respective skew body diagonals to them, respectively, then find

$$\frac{\left(\frac{1}{l^2} + \frac{1}{m^2} + \frac{1}{n^2}\right)}{\left(\frac{1}{OA^2} + \frac{1}{OB^2} + \frac{1}{OC^2}\right)}$$

Ans. 2

Sol. Let $l = m = n = \frac{1}{\sqrt{2}}$

Q62. If $\vec{V}_1 = \hat{i} + \hat{j} + \hat{k}$; $\vec{V}_2 = a\hat{i} + b\hat{j} + c\hat{k}$ where $a, b, c \in \{-2, -1, 0, 1, 2\}$. If p is the number of non zero vectors $\vec{V}_2$ which are perpendicular to $\vec{V}_1$. Then sum of digits of p is _____.

Ans. 9

Sol. We have

$$\vec{V}_1 \cdot \vec{V}_2 = a + b + c = 0$$

but $a, b, c \in \{-2, -1, 0, 1, 2\}$

now (i) if $a = 1, b = -1, c = 0$, number = $3! = 6$

(ii) if $a = 2, b = -2, c = 0$, number = $3! = 6$

(iii) if $a = 1, b = 1, c = -2$, number = $\frac{3!}{2!} = 3$

(iv) if $a = -1, b = -1, c = 2$, number = $\frac{3!}{2!} = 3$

$\therefore$ Total = 18

Q63. Let P and Q are two points in xy-plane on the curve $y = x^7 - 2x^5 + 5x^3 + 8x + 5$ such that $\vec{OP} \cdot \hat{i} = 2$ and $\vec{OQ} \cdot \hat{i} = -2$ and the magnitude of $\vec{OP} + \vec{OQ} = 2M$, (where O is origin) then find the value of M _____.

Ans. 5

Sol. Sol. Let P and Q be (x_1, y_1) and (x_2, y_2)

$$\therefore \vec{OP} \cdot \hat{i} = x_1 = 2 \text{ and } \vec{OQ} \cdot \hat{i} = x_2 = -2$$

$$\text{Let } y = f(x) = x^7 - 2x^5 + 5x^3 + 8x + 5$$

$$\therefore y_1 = f(x_1) = f(2) \text{ and } y_2 = f(x_2) = f(-2)$$

$$\begin{aligned} \therefore \vec{OP} + \vec{OQ} &= x_1\hat{i} + y_1\hat{j} + x_2\hat{i} + y_2\hat{j} \\ &= (f(2) + f(-2))\hat{j} \end{aligned}$$

Q64. If area of a triangular face BCD of a regular tetrahedron ABCD is $4\sqrt{3}$ sq. units, then find area of a triangle whose two sides are represented by vectors $\vec{AB}$ and $\vec{CD}$ _____.

Ans. 8

Sol. Sol. Let side be a

$$\begin{aligned} \frac{1}{2}a^2 \sin 60^\circ &= \frac{4\sqrt{3}}{2} \\ a^2 &= 16 \end{aligned}$$

$$\vec{AB} \cdot \vec{CD} = 0$$

$$\begin{aligned}\text{So area} &= \frac{1}{2}a^2 \sin 90^\circ \\ &= \frac{a^2}{2} = 8\end{aligned}$$

- Q65. If $\vec{a}, \vec{b}$ are two unit vectors and $\vec{c}$ is such that $\vec{c} = \vec{a} \times \vec{c} + \vec{b}$, then the maximum value of $[\vec{a}\vec{b}\vec{c}]$ is $\frac{p}{q}$ where p and q are co-prime natural numbers, then find the value of $(p + q)$ _____.

Ans. 3

Sol. $\vec{c} = \vec{a} \times \vec{c} + \vec{b}$
 $\Rightarrow |\vec{c} - \vec{b}|$
 $= |\vec{a} \times \vec{c}|$
 $\Rightarrow c^2 + 1 - 2\vec{b} \cdot \vec{c}$, where $\theta = \vec{a} \wedge \vec{c}$
 $= c^2 \sin^2 \theta$
 $\Rightarrow 2\vec{b} \cdot \vec{c}$
 $= c^2 \cos^2 \theta + 1$
 $\Rightarrow 2\vec{b} \cdot (\vec{a} \times \vec{c} + \vec{b})$
 $= c^2 \cos^2 \theta + 1$
 $\Rightarrow -2[\vec{a}\vec{b}\vec{c}] + 2 = c^2 \cos^2 \theta + 1$
 $\Rightarrow 2[\vec{a}\vec{b}\vec{c}]$
 $= 1 - c^2 \cos^2 \theta + 1$
 $\Rightarrow [\vec{a}\vec{b}\vec{c}]$
 $\leq 1/2$

- Q66. If $[\vec{a}\vec{b}\vec{c}] = 2$, and $\vec{d}$ is unit vector then find the value of $|(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{b} \times \vec{c}) \times (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \times (\vec{b} \times \vec{d})|$.

Ans. 4

Sol. Let $\vec{d} = \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c}$
 Now $\vec{d} \cdot \vec{b} \times \vec{c} = 2\alpha$
 $\vec{d} \cdot \vec{c} \times \vec{a} = 2\beta$
 $\vec{d} \cdot \vec{a} \times \vec{b} = 2\gamma \quad \therefore [\vec{d}\vec{b}\vec{c}]\vec{a} + [\vec{d}\vec{c}\vec{a}]\vec{b} + [\vec{d}\vec{a}\vec{b}]\vec{c}$
 $= 2\vec{d}$
 Now, $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{b} \times \vec{c}) \times (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \times (\vec{b} \times \vec{d})$
 $= [\vec{a}\vec{b}\vec{d}]\vec{c} - 2\vec{d} + [\vec{b}\vec{c}\vec{d}]\vec{a} - 2\vec{d} + [\vec{c}\vec{a}\vec{d}]\vec{b} - 2\vec{d}$
 $= -4\vec{d}$

- Q67. Let $\vec{a} = x^2\hat{i} - 3\hat{j} + (x+3)\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - (x-3)\hat{k}$ be two vectors such that $|\vec{a}| = |\vec{b}|$. If $\vec{v}_1 = 2\vec{a} + 3\vec{b}$ and $\vec{v}_2 = 3\vec{a} - 2\vec{b}$ and $I = \int_1^3 (\vec{v}_1 \wedge \vec{v}_2) dx$, then find the value of $[I]$ _____.
 [Note: $[k]$ denotes greatest integer function less than or equal to k .]

Ans. 3

Sol. $\vec{a} \cdot \vec{b} = (x^2\hat{i} + 3\hat{j} + (x+3)\hat{k}) \cdot (\hat{i} + 3\hat{j} - (x-3)\hat{k})$
 $= x^2 - 9 - (x^2 - 9) = 0$
 $\Rightarrow \vec{a} \perp \vec{b}$ and $|\vec{a}| \perp |\vec{b}|$

$$\vec{v}_1 \cdot \vec{v}_2 = (2\vec{a} + 3\vec{b}) \cdot (3\vec{a} - 2\vec{b})$$

$$6a^2 - 6b^2 - 4\vec{a} \cdot \vec{b} + 9\vec{b} \cdot \vec{a} = 0$$

$$\therefore \vec{v}_1 \wedge \vec{v}_2 = \frac{\pi}{2}$$

$$I = \int_1^3 (\vec{v}_1 \wedge \vec{v}_2) dx$$

$$\text{Now, } = \int_1^3 \frac{\pi}{2} dx$$

$$= \frac{\pi}{2} (3 - 1)$$

$$= \pi$$

$$\therefore [I] = 3$$

Q68. $\vec{p} = 3^{(\sin x + \tan x)} \hat{i} + 3^{(\tan x - 1)} \hat{j} - 3^{\sin x} \hat{k}$ and $\vec{q} = 3^{(\sin x - 1)} \hat{i} + (3^{\sin x + \tan x}) \hat{j} + (3^{\tan x}) \hat{k}$. If sum of all values of x in $[0, 50\pi]$ for which $\vec{p} \cdot \vec{q}$ is minimum is S , then find the value of $\left(\frac{S}{255\pi}\right)$.

Ans. 5

$$\text{Sol. } \vec{p} \cdot \vec{q} = 3^{2\sin x + \tan x - 1} + 3^{\sin x + 2\tan x - 1} - 3^{\sin x + \tan x}$$

$$3\vec{p} \cdot \vec{q} = 3^{2\sin x + \tan x} + 3^{\sin x + 2\tan x} - 3 \cdot 3^{\sin x + \tan x}$$

$$3\vec{p} \cdot \vec{q} + 1 = \left(3^{\frac{2\sin x + \tan x}{3}}\right)^3 + \left(3^{\frac{\sin x + 2\tan x}{3}}\right)^3 + 1^3 - 3 \cdot \left(3^{\frac{2\sin x + \tan x}{3}}\right) \cdot \left(3^{\frac{\sin x + 2\tan x}{3}}\right) \cdot 1$$

If a, b, c are positive then $a^3 + b^3 + c^3 - 3abc \geq 0$ and equality holds when $a = b = c$

$$\therefore \sin x = \tan x = 0 \Rightarrow x = n\pi$$

$$S = 0 + \pi + 2\pi + \dots + 50\pi$$

$$= \frac{\pi(50 \times 51)}{2}$$

$$\therefore \frac{S}{255\pi} = 5$$

Q69. Let a vector $\vec{V}_1 = \vec{OP}$, where O is origin and P is $(2, 0, 2)$. Vector $\vec{V}_1$ is rotated about origin by an angle of $\frac{\pi}{3}$ in a plane which is perpendicular to XOZ plane to get vector $\vec{V}_2$, where $\vec{V}_2 \cdot \hat{j} > 0$. Again $\vec{V}_1$ is rotated about origin by right angle in XOZ plane to get vector $\vec{V}_3$. If V is volume of tetrahedron whose coterminal edges are $\vec{V}_1, \vec{V}_2$ and $\vec{V}_3$ then value of $\frac{3V}{\sqrt{6}}$ is _____.

Ans. 4

Sol. Equation of plane perpendicular to XOZ plane

$$\text{Is } P: x - z = 0$$

$\vec{V}_2$ is coplanar with $\vec{V}_1$ and $\hat{j}$

$$\Rightarrow \vec{V}_1 = x\vec{V}_1 + y\hat{j}$$

$$\Rightarrow \vec{V}_2 = x(2\hat{i} + 2\hat{k}) + y\hat{j}$$

$$\therefore |\vec{V}_2| = |\vec{V}_1| = 2\sqrt{2}$$

$$\therefore 4x^2 + y^2 + 4x^2 = 8 \quad (1)$$

$$\text{Now, } \vec{V}_2 \cdot \vec{V}_1 = |\vec{V}_2||\vec{V}_1| \cos \frac{\pi}{3} = 4$$

$$4 = (x(2\hat{i} + 2\hat{k}) + y\hat{j}) \cdot (2\hat{i} + 2\hat{k})$$

$$\begin{aligned}
4 = 8x &\Rightarrow x = \frac{1}{2} \Rightarrow y = \pm\sqrt{6} \\
&\Rightarrow \vec{V}_2 \\
&= \hat{i} + \sqrt{6}\hat{j} + \hat{k} (\because \vec{V}_2 \cdot \hat{j} > 0) \\
\text{Now, } \vec{V}_3 &= \vec{V}_1 + x\hat{i} \\
\therefore \vec{V}_3 \cdot \vec{V}_1 &= 0 \Rightarrow 8 + 2x = 0 \Rightarrow x = -4 \\
&\Rightarrow \vec{V}_3 = 2\hat{i} + 2\hat{k} - 4\hat{j} \\
\vec{V}_3 &= -2\hat{i} + 2\hat{k} \\
&\Rightarrow \text{volume of tetrahedron } V = \left| \frac{1}{6} \begin{vmatrix} 2 & 0 & 2 \\ 1 & \sqrt{6} & 1 \\ -2 & 0 & 2 \end{vmatrix} \right| \\
&\Rightarrow V = \frac{1}{6} 8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}} = 4
\end{aligned}$$

Q70. Let 'n' represents the number of Lattice points (x, y, z all being integers) inside the tetrahedron (not on the surface) having vertices (0, 0, 0), (8, 0, 0), (0, 8, 0) and (0, 0, 8) then the value of $\left(\frac{n}{7}\right)$ is _____.

Ans. 5

Sol. $x + y + z < 8$, $x, y, z \geq 1$
 $\Rightarrow x + y + z + w = 8$
 $w \geq 1$
 $\Rightarrow$ No of solutions $= n = {}^7C_3 = 35$

Q71. Consider two straight lines $L_1 : \frac{x-2}{-1} = \frac{y-2}{1} = \frac{z-4}{2}$ and $L_2 : \frac{x-0}{1} = \frac{y+1}{2} = \frac{z-3}{-2}$. Let P and Q be the variable on L_1 and L_2 respectively and point M divides PQ in the ratio 2 : 3 internally. If point M moves in a plane $ax + bz = 6$, then value of $a + b$ is _____.

Ans. 3

Sol. $L_1 : \frac{x-2}{-1} = \frac{y-2}{1} = \frac{z-4}{2}$ and $L_2 : \frac{x-0}{1} = \frac{y+1}{2} = \frac{z-3}{-2}$

Using section formula and eliminating

λ, μ we get $w + 2\mu = 6$

$M(u, v, w)$ moves on

$2x + z = 6$

Q72. If $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is 60° . If $p_i, q_i, r_i; i = 1, 2, 3$ are scalars satisfying $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p_1\vec{a} + q_1\vec{b} + r_1\vec{c}$, $\vec{b} \times \vec{c} + \vec{c} \times \vec{a} = p_2\vec{a} + q_2\vec{b} + r_2\vec{c}$ and $\vec{c} \times \vec{a} + \vec{a} \times \vec{b} = p_3\vec{a} + q_3\vec{b} + r_3\vec{c}$ then

$$\begin{vmatrix} p_1 & q_1 & r_1 \\ p_2 & q_2 & r_2 \\ p_3 & q_3 & r_3 \end{vmatrix} = \lambda \begin{vmatrix} p_1 + q_1 & q_1 & r_1 \\ p_2 - r_2 & q_2 & r_2 \\ p_3 + 2q_3 & q_3 & r_3 \end{vmatrix}$$

The value of λ is equal to _____.

Ans. 1

Sol. 

Q73. A straight line cuts the sides AB, AC and AD of a parallelogram ABCD at points P, Q and R respectively. If $\vec{AP} = \lambda_1 \vec{AB}$, $\vec{AR} = \lambda_2 \vec{AD}$, $\vec{AQ} = \frac{\lambda_3}{2} \vec{AC}$, where $\lambda_1, \lambda_2, \lambda_3$ are scalars then prove that $\lambda_1, \lambda_2, \lambda_3$ are in H.P.

Ans. -

Sol. 

Q74. The radical centre of the following set of circles $x^2 + y^2 - 3x - 6y + 14 = 0$; $x^2 + y^2 - x - 4y + 8 = 0$, $x^2 + y^2 + 2x - 6y + 9 = 0$ is (α, β) . Find the equations of the straight line passing through the point $(\alpha, \beta, 3)$ to intersect the straight line $x + 1 = 2(y - 2) = z + 4$ and parallel to the plane $x + 5y + 4z = 0$.

Ans. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

Sol. $S_1 - S_2 = 0$

$$2x + 2y = 6$$

$$x + y = 3 \dots (1)$$

$$S_2 - S_3 = 0$$

$$3x - 2y + 1 = 0 \dots (2)$$

From eq. (1) & (2)

Radical center (1, 2)

Point P(1, 2, 3) & Line $\frac{x+1}{2} = \frac{y-2}{1} = \frac{z+4}{2} = \lambda$

Q $(2\lambda - 1, \lambda + 1, 2\lambda - 4)$

$$\vec{PQ} \cdot \vec{n} = 0$$

$$1(2\lambda - 2) + 5\lambda + 4(2\lambda - 7) = 0$$

$$15\lambda = 30$$

$$\lambda = 2$$

Q $(3, 4, 0) \Rightarrow$ Line $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

Q75. $\hat{W}$ is a unit vector perpendicular to the plane of two other unit vectors $\hat{u}$ and $\hat{v}$. Three unit vectors $\hat{x}, \hat{y}$ and $\hat{z}$ are along the angle bisectors of $\hat{u}, \hat{v}$; $\hat{v}, \hat{w}$ and $\hat{w}, \hat{u}$ respectively. If $[\hat{x} \times \hat{y} \hat{y} \times \hat{z} \hat{z} \times \hat{x}] = \frac{1}{2}$ then find the angle between $\hat{u}$ and $\hat{v}$.

Ans. $\pi/2$

Sol. Let angle between $\hat{u}$ and $\hat{v}$ be α then

$$\hat{x} = \frac{\hat{u} + \hat{v}}{2 \cos \frac{\alpha}{2}}, \hat{y}$$

$$= \frac{\hat{v} + \hat{w}}{\sqrt{2}}$$

$$\text{and } \hat{z} = \frac{\hat{u} + \hat{w}}{\sqrt{2}}$$

$$\begin{aligned}
&\Rightarrow [\hat{x}\hat{y}\hat{z}] \\
&= \frac{1}{4 \cos \frac{\alpha}{2}} [\hat{u} + \hat{v} \hat{v} + \hat{w} \hat{u} + \hat{w}] \\
&= \frac{2}{4 \cos \frac{\alpha}{2}} [\hat{u}\hat{v}\hat{w}]
\end{aligned}$$

Also

$$\begin{aligned}
&= [\hat{x}\hat{y}\hat{z}]^2 \\
&= \left(\frac{[\hat{u}\hat{v}\hat{w}]}{2 \cos \alpha/2} \right)^2 \\
&= \left(\frac{\sin \alpha}{2 \cos \alpha/2} \right)^2 \\
&= \frac{1}{2} \\
\Rightarrow \alpha &= \frac{\pi}{2}
\end{aligned}$$

Q76. The position vectors of the vertices A, B and C of a triangle are $\hat{i} + \hat{j}$, $\hat{j} + \hat{k}$ and $\hat{i} + \hat{k}$ respectively. Find a unit vector $\vec{r}$ lying in the plane of $\triangle ABC$ and perpendicular to IA, where I is the incentre of the triangle _____.

Ans. $\vec{r} = \pm \frac{1}{2}(\hat{i} - \hat{j})$

Sol. Since $|\overrightarrow{AB}| = |\overrightarrow{BC}| = |\overrightarrow{CA}|$, incentre is same as circumcentre, and hence IA is perpendicular to BC $\Rightarrow \vec{r}$ is parallel to BC $\Rightarrow \vec{r} = \lambda(\hat{i} - \hat{j})$.
Hence unit vector $\vec{r} = \pm \frac{1}{2}(\hat{i} - \hat{j})$

Q77. Four points with position vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are coplanar such that $(\sin \alpha)\vec{a} + (2 \sin 2\beta)\vec{b} + (3 \sin 3\gamma)\vec{c} - \vec{d} = 0$. Then, the least value of the expression $\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma$ is $\frac{p}{q}$, where p and q are coprime, then find $(p + q)$.

Ans. 15

Sol. For four coplanar points with position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$, the vector $\vec{d}$ can be expressed as an affine combination of $\vec{a}, \vec{b}, \vec{c}$.

The given relation is $(\sin \alpha)\vec{a} + (2 \sin 2\beta)\vec{b} + (3 \sin 3\gamma)\vec{c} - \vec{d} = 0$.

Thus $\vec{d} = (\sin \alpha)\vec{a} + (2 \sin 2\beta)\vec{b} + (3 \sin 3\gamma)\vec{c}$.

For this to be an affine combination of $\vec{a}, \vec{b}, \vec{c}$, the sum of its coefficients must be 1.

Therefore, $\sin \alpha + 2 \sin 2\beta + 3 \sin 3\gamma = 1$.

Now apply the Cauchy-Schwarz inequality to the vectors $(1, 2, 3)$ and $(\sin \alpha, \sin 2\beta, \sin 3\gamma)$.

We get $|\sin \alpha + 2 \sin 2\beta + 3 \sin 3\gamma| \leq \sqrt{1^2 + 2^2 + 3^2} \sqrt{\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma}$.

Using $\sin \alpha + 2 \sin 2\beta + 3 \sin 3\gamma = 1$, this becomes $1 \leq \sqrt{14} \sqrt{\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma}$.

Squaring both sides gives $\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma \geq \frac{1}{14}$.

Equality holds when $(\sin \alpha, \sin 2\beta, \sin 3\gamma) = k(1, 2, 3)$ for some k .

Then $k + 2(2k) + 3(3k) = 1$, so $14k = 1$, hence $k = \frac{1}{14}$.

This is attainable because $\frac{1}{14}, \frac{2}{14}, \frac{3}{14}$ all lie in $[-1, 1]$.

Thus the least value is $\frac{1}{14}$.

So $p = 1$ and $q = 14$.

Therefore, $p + q = 1 + 14 = 15$.

Q78. Match the following column :

Column-I

Column-II

(a) If the angle between the plane $x - 3y + 2z = 1$ and the line $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-1}{-3}$ is θ , then the value of $|\operatorname{cosec}\theta|$ is (p) 2

(b) If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar and $[\vec{a} \vec{b} \vec{c}] = \frac{4}{7}$, then $[2\vec{a} - \vec{b} \quad 2\vec{b} - \vec{c} \quad 2\vec{c} - \vec{a}]$ is (q) 8

(c) Let $\vec{r} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$, where $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors. If $\vec{r}$ is perpendicular to $(\vec{a} + \vec{b} + \vec{c})$ and the minimum value of $(x^2 + y^2)$ is $\frac{k\pi^2}{4}$, then 'k' is (r) 4

(d) If image of the point $(1, 0, 1)$ in the plane $x - y - z = 1$ is (a, b, c) then $3(a - b + c)$ is equal to (s) 5

(t) an odd integer

Choose the correct option :

- (A) (a) $\rightarrow$ (p); (b) $\rightarrow$ (r); (c) $\rightarrow$ (s,t); (d) $\rightarrow$ (q) (B) (a) $\rightarrow$ (q); (b) $\rightarrow$ (s); (c) $\rightarrow$ (p,r); (d) $\rightarrow$ (p)
(C) (a) $\rightarrow$ (r); (b) $\rightarrow$ (t); (c) $\rightarrow$ (p,q); (d) $\rightarrow$ (r) (D) (a) $\rightarrow$ (s); (b) $\rightarrow$ (p); (c) $\rightarrow$ (q,t); (d) $\rightarrow$ (p)

Ans. A

Sol. We evaluate each matching item in Column-I.

(a) The normal vector to the plane is $\vec{n} = (1, -3, 2)$ and the direction vector of the line is $\vec{d} = (2, 1, -3)$.

For the angle θ between a line and a plane, $\sin \theta = \frac{|\vec{n} \cdot \vec{d}|}{|\vec{n}||\vec{d}|}$.

Here $\vec{n} \cdot \vec{d} = 2 - 3 - 6 = -7$, so $|\vec{n} \cdot \vec{d}| = 7$.

Also $|\vec{n}| = \sqrt{14}$ and $|\vec{d}| = \sqrt{14}$, hence $\sin \theta = \frac{7}{14} = \frac{1}{2}$.

Therefore $|\operatorname{cosec} \theta| = 2$, so (a) matches (p).

(b) Let $\Delta = [\vec{a} \ \vec{b} \ \vec{c}] = \frac{4}{7}$.

The scalar triple product transforms as $[2\vec{a} - \vec{b} \ 2\vec{b} - \vec{c} \ 2\vec{c} - \vec{a}] = \det \begin{bmatrix} 2 & -1 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & 2 \end{bmatrix} \Delta$.

Computing the determinant gives $2(4 - 0) + (-1)(0 - 1) = 8 - 1 = 7$.

Thus the required triple product is $7\Delta = 7 \cdot \frac{4}{7} = 4$, so (b) matches (r).

(c) Since $\vec{r}$ is perpendicular to $\vec{a} + \vec{b} + \vec{c}$, we have $\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$.

Using cyclic scalar triple products, this becomes $\Delta(\sin x + \cos y + 2) = 0$.

Since $\Delta \neq 0$, we get $\sin x + \cos y = -2$.

This forces $\sin x = -1$ and $\cos y = -1$.

The minimum occurs at $x = -\frac{\pi}{2}$ and $y = \pm\pi$, giving $x^2 + y^2 = \frac{\pi^2}{4} + \pi^2 = \frac{5\pi^2}{4}$.

Thus $k = 5$, so (c) matches (s); since 5 is also an odd integer, it matches (t) as well.

(d) The reflection formula is $Q = P - 2 \frac{F(P)}{|\vec{n}|^2} \vec{n}$, where $F = x - y - z - 1$.

Here $P = (1, 0, 1)$, $\vec{n} = (1, -1, -1)$, and $F(P) = -1$, so $|\vec{n}|^2 = 3$.

Thus $Q = (1, 0, 1) + \frac{2}{3}(1, -1, -1) = \left(\frac{5}{3}, -\frac{2}{3}, \frac{1}{3}\right)$.

So $a = \frac{5}{3}$, $b = -\frac{2}{3}$, $c = \frac{1}{3}$, giving $a - b + c = \frac{8}{3}$.

Therefore $3(a - b + c) = 3 \cdot \frac{8}{3} = 8$, so (d) matches (q).

Hence the matching is (a) $\rightarrow$ (p); (b) $\rightarrow$ (r); (c) $\rightarrow$ (s,t); (d) $\rightarrow$ (q).

So the correct option is **A**.

Q79. Match the following column :

Column-I

Column-II

(a) Let volume of a tetrahedron ABCD is $\frac{81}{2}$ cube unit and volume of parallelepiped

whose three coterminous edges are segments joining centroid of any face of tetrahedron with centroids of its other three faces is V cubic unit, then V is

(p) 9

(b) If image of the point $(1, 0, 1)$ in the plane $x - y - z = 1$ is (a, b, c) then $3(a - b + c)$ is equal to

(q) 8

(c) Locus of all the points which are at a distance of 3 units from the line

$\vec{r} = \lambda(i + j + k)$ is given by $x^2 + y^2 + z^2 - xy - yz - xz = \frac{k}{2}$, then find $\frac{k}{9}$

(r) 3

(d) $\vec{a}, \vec{b}$ and $\vec{c}$ are such that $|\vec{a}| = \sqrt{3}, |\vec{b}| = 2$ and $|\vec{c}| = \sqrt{6}$. If $\vec{a} \cdot \vec{b} < 0; \vec{c}$ is

perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{a}, \vec{b}, \vec{c}$ form the coterminous edges of a tetrahedron of unit volume, then angle between $\vec{a}$ and $\vec{b}$ is $\frac{p\pi}{q}$, (where p, q are coprime numbers),

(s) 7

where $(p + q)$ is

Choose the correct option :

(A) (a) p; (b) q; (c) r; (d) s (B) (a) s; (b) r; (c) q; (d) p (C) (a) p; (b) s; (c) r; (d) q

(D) (a) p; (b) q; (c) s; (d) r

Ans. A

Sol. We evaluate each item in Column-I and match it with the corresponding value in Column-II.

(a) Let the tetrahedron have vertices A, B, C, D, and let its volume be T .

Take the centroid of face ABC and join it to the centroids of faces ABD, ACD, and BCD.

The three edge vectors of the parallelepiped are $\frac{D-C}{3}, \frac{D-B}{3}, \frac{D-A}{3}$.

$$\begin{aligned} \text{Hence } V &= \left| \det \left(\frac{D-C}{3}, \frac{D-B}{3}, \frac{D-A}{3} \right) \right| \\ &= \frac{1}{27} |\det(D-C, D-B, D-A)| \end{aligned}$$

Taking A as origin and writing $B-A = b, C-A = c, D-A = d$, this determinant has magnitude $|\det(b, c, d)|$.

Since $T = \frac{1}{6} |\det(b, c, d)|$, we get $V = \frac{1}{27} \cdot 6T = \frac{2}{9}T$.

Given $T = \frac{81}{2}$, $V = \frac{2}{9} \cdot \frac{81}{2} = 9$, so (a) matches (p).

(b) The plane is $x - y - z = 1$, so its normal is $n = (1, -1, -1)$ and $\|n\|^2 = 3$.

For $P = (1, 0, 1)$, the plane expression gives $1 - 0 - 1 - 1 = -1$.

The reflection formula gives $P' = P - 2 \frac{F(P)}{\|n\|^2} n = P + \frac{2}{3}n$.

$$\begin{aligned} P' &= (1, 0, 1) + \frac{2}{3}(1, -1, -1) \\ \text{Thus } &= \left(\frac{5}{3}, -\frac{2}{3}, \frac{1}{3} \right) \end{aligned}$$

$$\begin{aligned} 3(a - b + c) &= 3 \left(\frac{5}{3} - \frac{2}{3} + \frac{1}{3} \right) \\ \text{So } &= 3 \cdot \frac{8}{3} = 8, \text{ so (b) matches (q).} \end{aligned}$$

(c) The line is $\vec{r} = \lambda(i + j + k)$, so its direction is $d = (1, 1, 1)$ and $|d|^2 = 3$.

For a point (x, y, z) at distance 3 from the line, $\frac{|\vec{r} \times \vec{d}|}{|\vec{d}|} = 3$.

Squaring: $|\vec{r} \times \vec{d}|^2 = 27$.

Using $|\vec{r} \times \vec{d}|^2 = |\vec{r}|^2 |\vec{d}|^2 - (\vec{r} \cdot \vec{d})^2$, we get $3(x^2 + y^2 + z^2) - (x + y + z)^2 = 27$.

Expanding and simplifying: $x^2 + y^2 + z^2 - xy - yz - zx = \frac{27}{2}$.

Comparing with $\frac{k}{2}$, $k = 27$, hence $\frac{k}{9} = 3$, so (c) matches (r).

(d) The tetrahedron volume is $\frac{1}{6} |[\vec{a}, \vec{b}, \vec{c}]| = 1$.

Since $\vec{c}$ is perpendicular to both $\vec{a}$ and $\vec{b}$, $|[\vec{a}, \vec{b}, \vec{c}]| = |\vec{a} \times \vec{b}| |\vec{c}|$.

Let θ be the angle between $\vec{a}$ and $\vec{b}$.

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

$$\text{Then } = \sqrt{3} \cdot 2 \sin \theta.$$

$$= 2\sqrt{3} \sin \theta$$

Thus $\frac{1}{6} (2\sqrt{3} \sin \theta)(\sqrt{6}) = 1$, i.e. $\sqrt{2} \sin \theta = 1$, so $\sin \theta = \frac{1}{\sqrt{2}}$.

Given $\vec{a} \cdot \vec{b} < 0$, we have $\cos \theta < 0$, so $\theta = \frac{3\pi}{4}$.

Therefore $p = 3$, $q = 4$, and $p + q = 7$, so (d) matches (s).

Hence the matching is $(a) \rightarrow (p)$, $(b) \rightarrow (q)$, $(c) \rightarrow (r)$, $(d) \rightarrow (s)$.

This corresponds to option A.

Q80. Match the following column :

Column-I

Column-II

(a) If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, then the value of $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$ (p) 9

equal to

(b) If points P, Q and R have position vectors, $\vec{r}_1 = 3\hat{i} - 2\hat{j} - \hat{k}$, $\vec{r}_2 = \hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{r}_3 = 2\hat{i} + \hat{j} - 2\hat{k}$ respectively, relative to an origin O, then distance of P from the plane OQR is (q) 28

(c) A line with direction ratios (2, 1, 2) intersects the lines $\vec{r} = -\hat{i} + \lambda(\hat{i} + \hat{j} - \hat{k})$ and $\vec{r} = -\hat{i} + \lambda(2\hat{i} + \hat{j} + \hat{k})$ at A and B, then (AB) is equal to (r) 16

(d) In a regular tetrahedron, the centres of the four faces are the vertices of a smaller tetrahedron. The ratio Of the volume of the smaller tetrahedron to that of the larger is (s) $\frac{3}{m/n}$, where m and n are relatively prime positive integers. The value of (m + n) is

(A) $(a) \rightarrow (p)$, $(b) \rightarrow (r)$, $(c) \rightarrow (q)$, $(d) \rightarrow (s)$ (B) $(a) \rightarrow (q)$, $(b) \rightarrow (p)$, $(c) \rightarrow (r)$, $(d) \rightarrow (s)$

(C) $(a) \rightarrow (s)$, $(b) \rightarrow (q)$, $(c) \rightarrow (p)$, $(d) \rightarrow (r)$ (D) $(a) \rightarrow (r)$, $(b) \rightarrow (s)$, $(c) \rightarrow (s)$, $(d) \rightarrow (q)$

Ans. D

Sol. mid point of AA' lie on plane $x - y - z = 1 \Rightarrow \lambda = \frac{2}{3}$

$$a = \frac{5}{3}, b = \frac{-2}{3}, c = \frac{1}{3} \Rightarrow 3(a - b + c) = 3\left(\frac{8}{3}\right) = 8$$

$$(R) \vec{QP} \cdot (\hat{i} + \hat{j} + \hat{k}) = 0 \Rightarrow (x - \lambda) + (y - \lambda) + (z - \lambda) = 0$$
$$\lambda = \frac{x + y + z}{3}$$

$$(PQ)^2 = 9 \Rightarrow (x - \lambda)^2 + (y - \lambda)^2 + (z - \lambda)^2 = 9$$

$$x^2 + y^2 + z^2 + 3\lambda^2 - 2\lambda \cdot 3\lambda = 9$$

$$\Rightarrow x^2 + y^2 + z^2 - 3\left(\frac{x + y + z}{3}\right)^2 = 9$$

$$\Rightarrow x^2 + y^2 + z^2 - xy - yz - zx = \frac{27}{2}$$

$$k = 27, \frac{k}{9} = 3$$

$$(S) \vec{C} = \lambda(\vec{a} \times \vec{b}) \text{ also } \left| \frac{1}{6} [\vec{a}\vec{b}\vec{c}] \right| = 1$$
$$\Rightarrow |[\vec{a}\vec{b}\vec{c}]| = 6$$

$$\Rightarrow |(\vec{a} \times \vec{b}) \cdot \vec{c}|$$
$$= 6$$

$$\Rightarrow \frac{|\vec{c}|^2}{|\lambda|}$$

$$= 6$$

$$\Rightarrow 6$$

$$= 6|\lambda|$$

$$\Rightarrow |\lambda|$$

$$= 1$$

$$\therefore |\vec{c}|^2 = \lambda^2 |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta$$

$$\Rightarrow 6 = 12 \sin^2 \theta = \theta = \frac{3\pi}{4} \quad (a \cdot b < 0) p + q = 7$$